RTA-SUM - State of Health Manager (SOH)

Design Document

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Contents

1 Introduction and Goals **5**
1.1 Introduction . . . 5
1.1.1 System Purpose . . . 5
1.1.2 Document Purpose . . . 5
1.2 Goals of the Architecture . . . 5
1.3 Stakeholders . . . 6
1.4 Definitions, Acronyms and Abbreviations . . . 6
1.5 Referenced Documents . . . 7

2 Architecture Constraints **8**
2.1 Environment . . . 8
2.2 Layers Definition . . . 8
2.3 Generic component interface . . . 9
2.3.1 Port Interfaces . . . 9
2.4 Other constraints and assumptions . . . 13

3 Design Overview **15**
3.1 Overview . . . 15
3.2 Memory Types . . . 16
3.3 Filters . . . 18
3.4 Reliability monitors . . . 20
3.4.1 Reset Reliability monitor . . . 20
3.4.2 Memory Status Reliability Monitor . . . 23
3.4.3 Communication Reliability Monitor . . . 27
3.4.4 Watchdog Manager Reliability Monitor . . . 32
3.4.5 Platform Reliability Monitor . . . 35
3.4.6 Data Provider . . . 35
3.5 RID 0x02C0 Clear SOH Logs . . . 36
3.6 DID Counters . . . 38
3.6.1 DID 0xF08F: SUM_SOH Reset Reliability Monitor . . . 38
3.6.2 DID 0xF090: SUM_SOH Memory Status Reliability Monitor . . . 38
3.6.3 DID 0xF091: SUM_SOH Platform Reliability Monitor . . . 39
3.6.4 DID 0xF092: SUM_SOH Communication Reliability Monitor . . . 40
3.6.5 DID 0xF093: SUM_SOH Watchdog Manager Reliability Monitor . . . 41
3.7 Configuration . . . 41
3.7.1 Included Containers . . . 41
3.7.2 SOHGeneral . . . 42
3.7.3 SOHConfigSet . . . 47

4 Runtime View (Dynamic View) **68**
4.1 Module Initialization . . . 68
4.2 SOH Main . . . 69
4.2.1 SOH Main FSM . . . 70
4.3 SOH Time management . . . 71
4.4 Tasks management . . . 71
4.4.1 Periodic Runnables . . . 71
4.4.2 Event-Triggered Runnables . . . 71
4.5 Failure handling strategy . . . 71

4.6 Critical timing constraints 71
4.7 Runtime Scenario 71
4.7.1 Definitions 71
4.7.2 Variables 71
4.7.3 Functions 71

5 Deployment View **72**
5.1 Resource Limits 72

1 Introduction and Goals

This document intends to describe the design of the SUM_SOH module. The scope of this document is to capture the implementation based on the requirements as defined in GB8002 AUTOSAR Standard Utility Module Specification. For a better process and higher quality design, we can consider this work as an extension to the before-mentioned document.

1.1 Introduction

This section lists some of the driving forces that software architects must consider in their decisions. In particular, we will first present the purpose of the system, and then we will describe the purpose of the document.

1.1.1 System Purpose

This document refers to a set of standard software compositions which are named GM Standardized Utility Modules (SUM). Use of the SUM modules allows for the standardized specification of behaviorial requirements and expectations concerning the interface to the software infrastructure (BSW Services). Hence, we can completely abstract functional software from the software infrastructure. We can use SUMs to communicate a set of standardized implementations of the interactions between SW-C and BSW.

The SUM_SOH is a software module in the application layer of the standardized architecture of AUTOSAR.

SUM_SOH implements the following functionalities:

- Reset reliability monitor
- Memory Status reliability monitor
- Platform reliability monitor
- Communication reliability monitor
- Watchdog Manager reliability monitor
- Data provider via DIDs

1.1.2 Document Purpose

This document introduces the key ideas of the architecture of SUM_SOH. Its intended audience comprises System Architects, Product Architects, Developers and Testers who need a rough understanding of the concepts of the product. The document describes the product and its functionality on the component level, the interaction and communication between those components and the integration of ETAS wide shared-components. The document does not explain the design of the components in detail.

1.2 Goals of the Architecture

The main goals of the architecture are:

1. **Configurability**: The module has to be designed with the highest possible level of configurability, to be easy to adapt it in the future to changes on the standard or even in some project requirements.
2. **Maintainability**: The architecture of the component shall be clearly defined and easy to understand, to guarantee the maximum possible level of maintainability.
3. **Performance**: The component shall be designed to maximize run-time execution speed as well as minimize memory consumption.
4. **Correctness of Implementation**: The Detailed Design shall follow the requirements listed in the Architecture document (GB8002).

1.3 Stakeholders

Stakeholders for SUM_SOH module are GM, Bosch, Systems engineers, Software architect, QA personnel.

1.4 Definitions, Acronyms and Abbreviations

Term/Abbreviation	Definition
API	Application Program Interface
BSW	Basic Software
BswM	Basic Software Mode Manager
CAN	Controller Area Network
Dcm	Diagnostic Communication Module
DID	Data IDentifier
DTC	Diagnostic Trouble Code
DIAG	Diagnostic Application Services
ECU	Electronic Control Unit
EcuM	ECU State Manager
ERRH	Error Handling Module
LIN	Local Interconnect Network
LNM	LIN Network Manager
NVRAM	Non Volatile Random Access Memory
NvM	NVRAM Manager
OS	Operating System
OAW	Open Architecture Ware
PMS	Power Mode Slave
PNC	Partial Network Coordinator
RTE	Run Time Environment
SOH	State Of Health Manager

Term/Abbreviation	Definition
SSM	Signal Status Monitor
SSC	Security Service Coordinator
SUM	Standard Utility Module
SUSD	Startup Shut down Coordinator
SW	Software
SW-C	Software Component
TCS	Timing Client Service

1.5 Referenced Documents

Document Name	Description	Version	Released
SDV-8002	AUTOSAR Standard Utility Modules Specification	27.27.180.3p0	04/04/2025
SDV-6001	Diagnostic Infrastructure Specification	27.27.180.3p0	04/09/2025
CYS5402	SecOC and SOME IP Deployment Requirements	27.27.180.3p0	03/04/2025

2 Architecture Constraints

This chapter lists the constraints of the architecture. For instance, due to those constraints, some alternatives may not be evaluated and some decisions have to be made in a certain direction. In particular, we will first describe the development environment. Second, we will identify the layers of the software architecture and locate SUM_SOH in one specific layer. Finally, we will present the generic component interface and we will conclude by evaluating other possible constraints and assumptions.

2.1 Environment

Tool Name	Version	Description
ISOLAR-AB	12.3.3BETA1	AUTOSAR Authoring and BSW Configuration tools.
RTA-BSW	12.3.3pr1	BSW automatic configuration and code generation.
RTA-RTE	12.4.0	RTE generator used for testing
RTA-OS	6.1.0	OS that is used for testing, as AUTOSAR OS
MemMapGen	12.3.0	MemMap header file generator.
Cantata	9.5	Used for functional testing

2.2 Layers Definition

The SUM_SOH is a software module in the application layer of the standardized architecture of AUTOSAR (Figure 2.1). It interacts with other RTA-SUM modules and components of the BSW. In the latter case, the RTE is the means through which the interaction takes place.

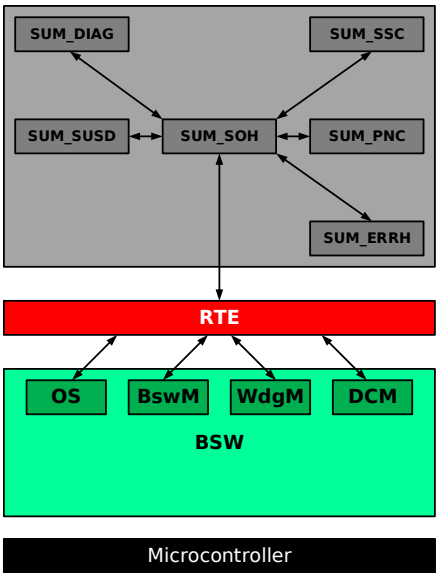


Figure 2.1: System layer overview.

2.3 Generic component interface

Standard Utility Modules are dependent on each other for some functions. The SUM_SOH dependencies require that ECUs deploying the SUM_SOH shall also deploy the following SUMs: SUM_DIAG, SUM_SUSD, SUM_SSC, SUM_PNC and SUM_ERRH. By means of the RTE, it also interacts with the OS, BswM, DCM and WdgM (Figure 2.2). .

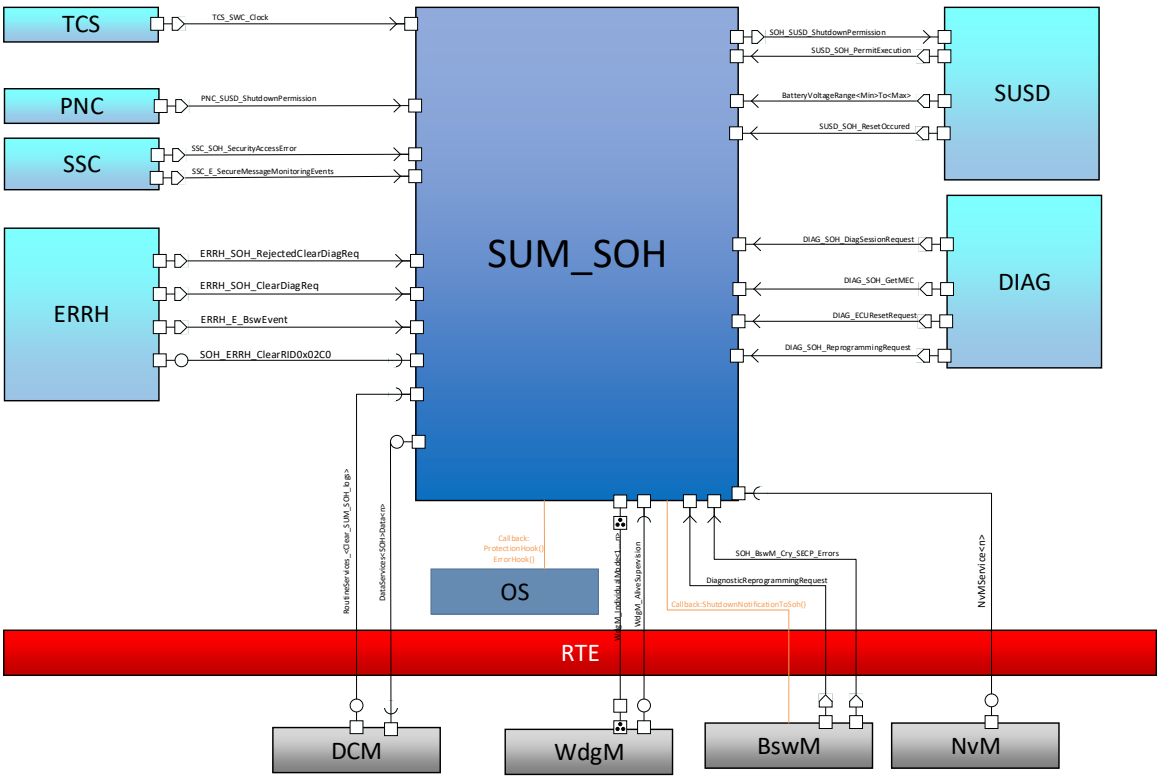


Figure 2.2: Generic Component Interface

Error detection based on host ECU type shall support BSW that fulfills the requirements of AUTOSAR_SWS_BasicSoftwareManager. BSW that fulfills the requirements of AUTOSAR_SWS_COM. BSW that fulfills the requirements of AUTOSAR_SWS_DiagnosticCommunicationManager. BSW that fulfills the requirements of AUTOSAR_SWS_DiagnosticEventManager. BSW that fulfills the requirements of AUTOSAR_SWS_WatchdogManager. BSW that fulfills the requirements of AUTOSAR_SWS_NVRAMManager. BSW that filfills the requirements of AUTOSAR_SWS_OS.

2.3.1 Port Interfaces

[$SDD_SOH 013]
All ECU types shall support Client / Server port interfaces. All ECU types shall support Sender / Receiver port interfaces.

[$Satisfies $SRS_SOH 016] [$Satisfies $SRS_SOH 228] [$Satisfies $SRS_SOH 229] [$Satisfies $SRS_SOH 230]
[$Satisfies $SRS_SOH 232] [$Satisfies $SRS_SOH 233] [$Satisfies $SRS_SOH 234] [$Satisfies $SRS_SOH 235]
[$Satisfies $SRS_SOH 243] [$Satisfies $SRS_SOH 248] [$Satisfies $SRS_SOH 249] [$Satisfies $SRS_SOH 250]
[$Satisfies $SRS_SOH 253] [$Satisfies $SRS_SOH 262]

Port Interface	Port Type	P-Port	R-Port	Description
ERRH_E_BswEvent	Sender / Re-ceiver	SUM_ERRH	SUM_SOH	Reception of DEM events which are routed from SUM_ERRH to the SUM_SOH for processing.
SUSD_SOH_PermitExecution	Sender / Re-ceiver	SUM_SUSD	SUM_SOH	Permit Execution interface to indicate that operating conditions are correct for the SUM_SOH to execute its functions.
PNC_SUSD_ShutdownPermission	Sender / Re-ceiver	SUM_PNC	SUM_SOH	Notification to the SUM_SOH of any active or not active partial networks.
SSC_E_SecureMessageMonitoringEvents	Sender / Re-ceiver	SUM_SSC	SUM_SOH	Notification of MAC failures along with their types.
DIAG_ECUResetRequest	Sender / Re-ceiver	SUM_DIAG	SUM_SOH	Notification of a service 0x11 tester-initiated request for ECU Reset (not programming).
SUSD_SOH_ResetOccured	Sender / Re-ceiver	SUM_SUSD	SUM_SOH	Notification to the SUM_SOH that a reset occurred. The type of reset is also communicated to the SUM_SOH. Type of resets may be delayed or immediate.

Port Interface	Port Type	P-Port	R-Port	Description
SSFM_ClearDiagReq	Sender / Receiver	SUM_SSFM	SUM_SOH	Notification to the SUM_SOH that a service 0x04 or a service 0x14 request was accepted or rejected.
DIAG_SOH_DiagSessionRequest	Sender / Receiver	SUM_DIAG	SUM_SOH	Notification to the SUM_SOH that a reprogramming request took place (service 0x10 subfunction 0x02).
SSC_SOH_SecurityAccessError	Sender / Receiver	SUM_SSC	SUM_SOH	Notification to the SUM_SOH of the occurrences of service 0x27 failures.
BatteryVoltage_DisableNVMWrite	Sender / Receiver	SUM_SUSD	SUM_SOH	Notification to the SUM_SOH from SUM_SUSD that vehicle battery voltage is too low to perform NVM writes.
DIAG_GetMEC	Sender / Receiver	SUM_DIAG	SUM_SOH	This interface is used by the SUM_DIAG to communicate to SUM_SOH to value of the MEC.
DataServices<SOH>Data<n>	Client / Server	DCM	SUM_SOH	To support the collection of Diagnostic DIDs that are part of the SUM_SOH functionalities.

Port Interface	Port Type	P-Port	R-Port	Description
DiagnosticReprogrammingRequest	Mode Switch	BswM	SUM_SOH	Notification of a request to shutdown to reset as part of programming to jump to boot.
SOH_WdgM_IndividualMode<1...n>	Mode Switch	WdgM	SUM_SOH	Notification of WdgM local supervision failures occurrence for all supervised entities excluding the SUM_SOH and the SUM_ERRH. These interfaces replace the SOH_BswM_WdgM _Individ-ualMode<1...n> interfaces described in GB8002.
SOH_WdgM_IndividualMode_Errh	Mode Switch	WdgM	SUM_SOH	Notification of the WdgM SUM_ERRH local supervision failures occurrences. This interface replaces the SOH_BsWM_WdgM _ERRHMode interface described in GB8002.
RP_ErrorStatus	Mode Switch	BswM	SOH	This interface is used by the BswM to notify SUM_SOH of SecP errors occurrences.
NvMService	Client / Server	NvM	SOH	The interface provides services for the management and maintenance of NV data related to SOH Metrics.
TCS_SWC_Clock	Client / Server	TCS	SOH	Provides Global Time for timestamp (unit 100ms).

Port Interface	Port Type	P-Port	R-Port	Description
RoutineServices_Clear_SUM_SOH_logs	Client / Server	DCM	SOH	The interface provides services for the clear of the SOH Metrics.
WdgM_LocalSupervision	Client / Server	WdgM	SUM_SOH	Notify the checkpoint reachement for implementing the Local Supervision.
SOH_ERRH_ClearRID0x02C0	Client / Server	SUM_ERRH	SUM_SOH	The interface provides service for starting clear, all the metrics related to SUM_SOH Local Supervision Detector implemented at ERRH side.
SOH_SUSD_ShutdownPermission	Sender / Re-ceiver	SUM_SOH	SUM_SUSD	Shutdown Permission interface to indicate that operating conditions are correct for the SUM_SOH to execute the shutdown.

2.4 Other constraints and assumptions

The ECU in which the SUM_SOH is implemented shall have a Secure cumulative NVM memory type, as defined in CTRS section 7.1.6, accessible to the SUM_SOH. The ECU in which the SUM_SOH is implemented shall have NonClearedRAM memory type accessible to the SUM_SOH. The ECU in which the SUM_SOH is implemented shall have an AdaptableNVM memory type accessible to the SUM_SOH.

The ECU initialization sequence shall be implemented such that any NVM needed for the SUM_SOH function is available prior to the SUM_SOH being initialized.

For ECUs that implement Memory Partitioning mechanisms, the SUM_SOH shall be implemented in the supervisory partition of the ECU software. Rationale: SUM_SOH must be in the trusted partition in order to interface with OS services. Supervisory partition is expected to include BSW Modules.

The SUM_SOH shall be implemented as a Supervised Entity of the Watchdog Manager.

For the SUM_SOH DIDs, the relative Dcm parameter *DcmDspDataUsePort* must be set to CLIENT_SERVER. Starting from this configuration, SUM_SOH will populate the DIDs values in a BIG_ENDIAN format as it is expected at the UDS layer (because the *DcmDspDataEndianness* has no effect if CLIENT_SERVER is selected for *DcmDspDataUsePort*).

3 Design Overview

This chapter will include the old Building Block and Runtime View.

This chapter is going to describe the static decomposition of the system into building blocks. It is going to show also the relationship between these blocks. This chapter describes the behavior and interaction of the system's building blocks as runtime elements.

3.1 Overview

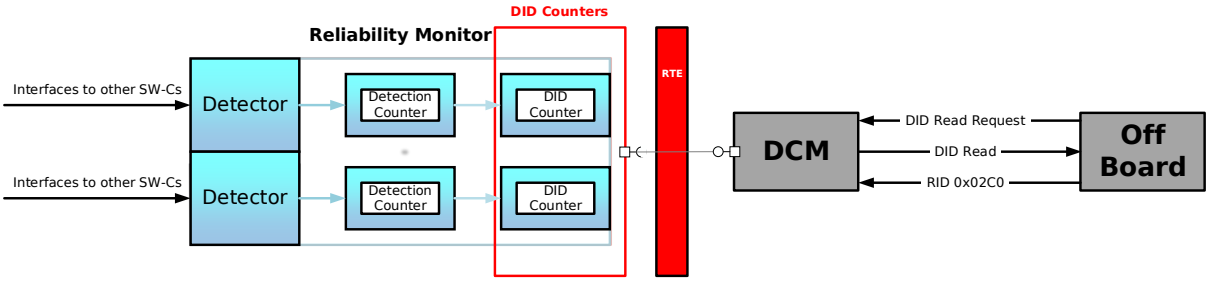


Figure 3.1: Example reliability monitor detectors, DID counters, DID read or RID execution

The SUM_SOH is defined to deploy functionalities needed to support monitoring and reporting of multiple facets of the operational health of the host ECU communication systems, hardware, operating system and software components.

The SUM_SOH design requires the use of software design elements called reliability monitors and detectors to support the implementation of infrastructure related functions (Figure 3.1).

The SUM_SOH reliability monitors are software functions that rely on one or more logically grouped internal and/or **external** detectors to filter and record the performance metrics of similar operating system constructs such as memory operation, network communication behavior and reset behavior (Figure 3.2).

The SUM_SOH detectors are software functions defined in **software performance specifications**. They are used for monitoring the performance of supervised/monitored operations, but can also be useful for functionalities other than those associated with reliability monitors.

The SUM_SOH requirements define detection counters used to maintain the raw count of specific errors or faults recorded by detectors.

Filtering of events based on defined calibration values is required for the SUM_SOH implementation, and to that end the requirements define multiple DID counters needed to maintain the filtered count of specific events.

Moreover, the SUM_SOH provides data regarding monitored events via DIDs allocated for the SUM_SOH and SUM_ERRH.

The SUM_SOH detectors can use one of two types of filtering:

- Threshold filtering. This type of filtering defines a threshold on the count of monitored events

as a calibration value to be used by the SUM_SOH. Once the defined threshold is reached the SUM_SOH will take the following actions.

- Threshold during a time window filtering. In this type of filtering, two calibration values are defined to be used by the SUM_SOH. The first is a threshold of the detection count of monitored events and the second is a threshold of the events detection time window. If the threshold of events is reached within the defined time window threshold, the SUM_SOH will take the following actions.

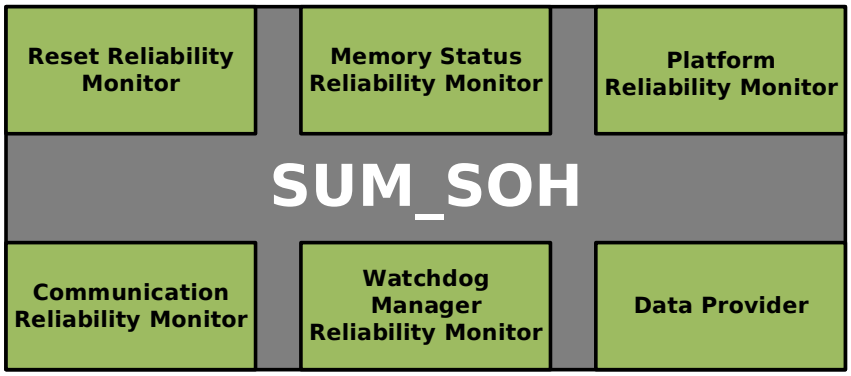


Figure 3.2: Inner structure of the system

3.2 Memory Types

[$SDD_SOH 019]
To store the SUM_SOH metrics four different types of memory are needed :

- RAM
- Non Cleared RAM
- Adaptable NVM
- Secure Cumulative NVM

Those types of memory shall be provided by the integrator. In partucular, for Adaptable NVM and Secure Cumulative NVM the ECU should provide an NvM module that meets the GM definition for those two types. Instead, for RAM and Non Cleared RAM types, the MemMap header file is generated using the MemMapGen generator present in RTA-CAR tool.
[$Satisfies $SRS_SOH 013] [$Satisfies $SRS_SOH 066] [$Satisfies $SRS_SOH 067] [$Satisfies $SRS_SOH 068]

[$SDD_SOH 058]
For those counters not located in RAM, if their values changed and the battery voltage is in range, their values are saved in the NvM.
[$Satisfies $SRS_SOH 027] [$Satisfies $SRS_SOH 265]

[$SDD_SOH 057]
The memory type where the SUM_SOH detection counters are located shall be assigned as defined in the following table 3.3.

[$Satisfies $SRS_SOH 029] [$Satisfies $SRS_SOH 047] [$Satisfies $SRS_SOH 066]

Detection Counter Name	Memory Type
SUM_SOH_ImmeReset_Cntr	NonClearedRAM
SUM_SOH_DelayedReset_Cntr	NonClearedRAM
SUM_SOH_DiagReset_Cntr	AdaptableNVM
SUM_SOH_StatNVM_Cntr	RAM
SUM_SOH_ActNVM_Cntr	RAM
SUM_SOH_PrivNVM_Cntr	RAM
SUM_SOH_ExcepOS_Cntr	NonClearedRAM
SUM_SOH_ServOS_Cntr	NonClearedRAM
SUM_SOH_StackOS_Cntr	NonClearedRAM
SUM_SOH_MemProtOS_Cntr	NonClearedRAM
SUM_SOH_TimeProtOS_Cntr	NonClearedRAM
SUM_SOH_ErrOS_Cntr	NonClearedRAM
SUM_SOH_ShutDown_Cntr	AdaptableNVM
SUM_SOH_ClearDiagInfo_Cntr	RAM
SUM_SOH_Reject_ClearDiag_Cntr	RAM
SUM_SOH_DiagSession_Cntr	Secure Cumulative NVM
SUM_SOH_Accept_DiagSession_Cntr	Secure Cumulative NVM
SUM_SOH_SecurityAccess_Cntr	Secure Cumulative NVM
SUM_SOH_MACfail_TenAct_Cntr	RAM
SUM_SOH_ReplayCount_Cntr	RAM
SUM_SOH_SecP_Cntr	RAM
SUM_SOH_LocalStatFail_Cntr	NonClearedRAM
SUM_SOH_WdgM_DEMErr_Cntr	NonClearedRAM
ERRH_SOHSupFail_Cntr	NonClearedRAM
SOH_ERRHSupFail_Cntr	NonClearedRAM

Figure 3.3: Detector memory types

3.3 Filters

There are two types of filter, the "Threshold" one checks if the number of events are above the threshold, the "Time window" insted checks if the number of events are above the threshold in a defined time span.

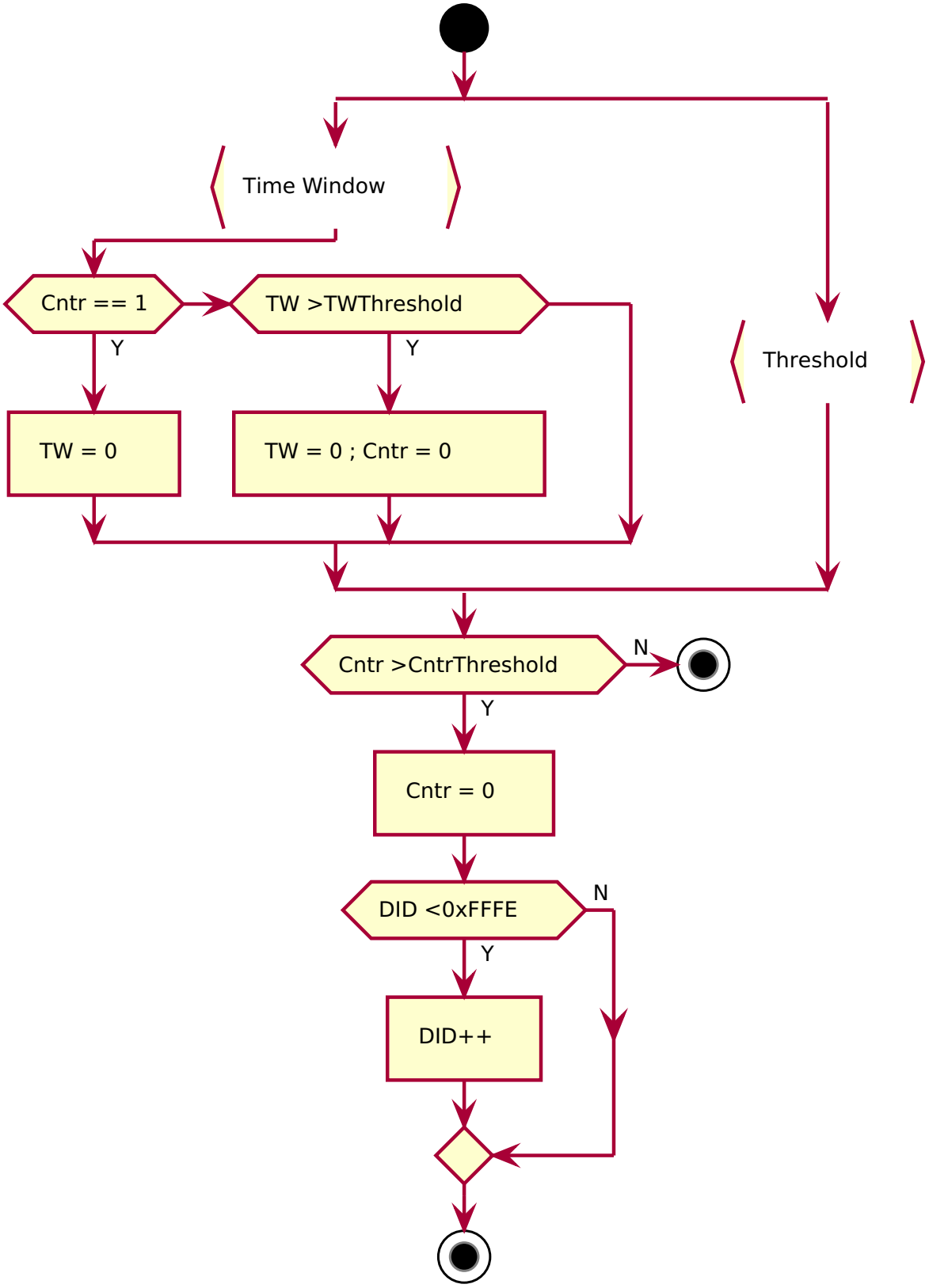


Diagram 3.1: SOH Filters

Threshold Filtering

[$SDD_SOH 011]
A 2-byte unsigned integer is used as detection counter to maintain a raw count of the detected events. Each detection counter shall reside in its respective memory and shall be initialized to a value of 0x0000 at ECU production. A 2-byte unsigned integer is used as DID counter to maintain a filtered count of the detected events. Each shall be stored in secure cumulative NVM. Each shall be initialized to a value of 0x0000 at ECU production.
[$Satisfies $SRS_SOH 028] [$Satisfies $SRS_SOH 029] [$Satisfies $SRS_SOH 030] [$Satisfies $SRS_SOH 033] [$Satisfies $SRS_SOH 041] [$Satisfies $SRS_SOH 042]

[$SDD_SOH 014]
Each detection counter will have a counter threshold calibration value associated with it
[$Satisfies $SRS_SOH 032]

[$SDD_SOH 015]
Each detection counter and DID counter shall be cleared to a value of 0x0000 during execution of RID 0x02C0 Clear_SUM_SOH_logs.
[$Satisfies $SRS_SOH 031] [$Satisfies $SRS_SOH 043]

[$SDD_SOH 020]
In the case where the detection counter reaches its threshold count calibration, the following steps SUM_SOH:

- Increments the DID counter by one.
- Resets the detection counter to 0x0000 and starts a new detection cycle

Also, if the DID counter reaches its maximum value of 0xFFFE before read time, that value is retained until the next execution of RID 0x02C0 Clear_SUM_SOH_logs.
[$Satisfies $SRS_SOH 044] [$Satisfies $SRS_SOH 045]

Threshold during a time window filtering

This feature is described in the Architecture Design document.

3.4 Reliability monitors

3.4.1 Reset Reliability monitor

[$SDD_SOH 025]
The scope of this monitor is to track the occurrences of three types of reset : Immediate, Delayed and Diagnostic resets.
[$Satisfies $SRS_SOH 071] [$Satisfies $SRS_SOH 072] [$Satisfies $SRS_SOH 079] [$Satisfies $SRS_SOH 086] [$Satisfies $SRS_SOH 022]

[$SDD_SOH 026]

Immediate Reset Detector

To handle the Immediate reset a Time Window filter is used with the following variables :

- Counter : SUM_SOH_ImmeReset_Cntr
- Counter Threshold : SUM_SOH_ImmeReset_Thresh
- Time Window : SUM_SOH_ImmeReset_TW
- Time Window Threshold : SUM_SOH_ImmeReset_TW_Thresh
- DID : SUM_SOH_DID_ImmeReset_Cntr

[$Satisfies $SRS_SOH 073] [$Satisfies $SRS_SOH 074] [$Satisfies $SRS_SOH 075] [$Satisfies $SRS_SOH 076] [$Satisfies $SRS_SOH 077] [$Satisfies $SRS_SOH 078] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 027]

Delayed Reset Detector

To handle the Delayed reset a Time Window filter is used with the following variables :

- Counter : SUM_SOH_DelayedReset_Cntr
- Counter Threshold : SUM_SOH_DelayedReset_Thresh
- Time Window : SUM_SOH_DelayedReset_TW
- Time Window Threshold : SUM_SOH_DelayedReset_TW_Thresh
- DID : SUM_SOH_DID_DelayedReset_Cntr

[$Satisfies $SRS_SOH 080] [$Satisfies $SRS_SOH 081] [$Satisfies $SRS_SOH 082] [$Satisfies $SRS_SOH 083] [$Satisfies $SRS_SOH 084] [$Satisfies $SRS_SOH 085] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 028]

Diagnostic Reset Detector

To handle the Diagnostic reset a Threshold filter is used with the following variables :

- Counter : SUM_SOH_DiagReset_Cntr
- Counter Threshold : SUM_SOH_DiagReset_Thresh
- DID : SUM_SOH_DID_DiagReset_Cntr

[$Satisfies $SRS_SOH 087] [$Satisfies $SRS_SOH 088] [$Satisfies $SRS_SOH 089] [$Satisfies $SRS_SOH 090] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 029]

The events about **Immediate** and **Delayed resets** are provided by SUM_SUSD.

The **`Runnable_SOHResetReliabilityMonitor_ImmeDelayedReset`** handles the type of reset through the interface `SUSD_SOH_ResetOccured` and increments of one unit the counter `SSUM_SOH_ImmeReset_Cntr` or `SSUM_SOH_DelayedReset_Cntr` based on the read value.

It calls the `SOH_Filter` on the activeted detector and if the DID has changed and if the battery voltage is in range, the DID is saved in the NvM.

[$Satisfies $SRS_SOH 020] [$Satisfies $SRS_SOH 235]

[$SDD_SOH 030]
The events about **Diagnostic reset** are provided by SUM_DIAG.

The **`Runnable_SOHResetReliabilityMonitor_DiagReset`** handles them through the interface `DIAG_ECUResetRequest` and increments of one unit the counter `SUM_SOH_DiagReset_Cntr`.

It calls the `SOH_Filter` on the activeted detector and if the DID or the Counter have changed and if the battery voltage is in range, the DID and the Counter are saved in the NvM.
[$Satisfies $SRS_SOH 029] [$Satisfies $SRS_SOH 047] [$Satisfies $SRS_SOH 027] [$Satisfies $SRS_SOH 265] [$Satisfies $SRS_SOH 086] [$Satisfies $SRS_SOH 246]

3.4.2 Memory Status Reliability Monitor

[$SDD_SOH 044]
The scope of this monitor is to track the occurrences of two types of faults : NvM and OS faults.
[$Satisfies $SRS_SOH 091]

[$SDD_SOH 045]

NVM Errors Detector

To handle the NVM faults, three different detectors based on a Time Window filter are used with the following variables :

Status Failure/Integrity

- Counter : SUM_SOH_StatNVM_Cntr
- Counter Threshold : SUM_SOH_StatNVM_Thresh
- Time Window : SUM_SOH_StatNVM_TW
- Time Window Threshold : SUM_SOH_StatNVM_TW_Thresh
- DID : SUM_SOH_DID_StatNVM_Cntr

Action Failure

- Counter : SUM_SOH_ActNVM_Cntr
- Counter Threshold : SUM_SOH_ActNVM_Thresh
- Time Window : SUM_SOH_ActNVM_TW
- Time Window Threshold : SUM_SOH_ActNVM_TW_Thresh
- DID : SUM_SOH_DID_ActNVM_Cntr

Privilege

- Counter : SUM_SOH_PrivNVM_Cntr
- Counter Threshold : SUM_SOH_PrivNVM_Thresh
- Time Window : SUM_SOH_PrivNVM_TW
- Time Window Threshold : SUM_SOH_PrivNVM_TW_Thresh
- DID : SUM_SOH_DID_PrivNVM_Cntr

[$Satisfies $SRS_SOH 091][$Satisfies $SRS_SOH 092][$Satisfies $SRS_SOH 093][$Satisfies $SRS_SOH 094]
[$Satisfies $SRS_SOH 096][$Satisfies $SRS_SOH 097][$Satisfies $SRS_SOH 098][$Satisfies $SRS_SOH 099]
[$Satisfies $SRS_SOH 100][$Satisfies $SRS_SOH 101][$Satisfies $SRS_SOH 102][$Satisfies $SRS_SOH 226]
[$Satisfies $SRS_SOH 227]

[$SDD_SOH 046]
The failure events are provided by SUM_ERRH.

The **`Runnable_SOH_ERRH_BswEvent`** handles the faults through the interface `ERRH_E_BswEvent` and, based on the read value, increments of one unit the related counter.
It calls the `SOH_Filter` on the activeted detector and if the DID has changed and if the battery voltage is in range, the DID is saved in the NvM.

The mapping between NVM Errors and detectors is the following :

Fault	Detector
NVM_E_HARDWARE	Status Failure/Integrity
NVM_E_QUEUE_OVERFLOW	Status Failure/Integrity
NVM_E_INTEGRITY_FAILED	Status Failure/Integrity
NVM_E_REQ_FAILED	Action Failure
NVM_E_WRONG_BLOCK_ID	Action Failure
NVM_E_VERIFY_FAILED	Action Failure
NVM_E_LOSS_OF_REDUNDANCY	Action Failure
NVM_E_WRITE_PROTECTED	Privilege

[$Satisfies $SRS_SOH 095]

[$SDD_SOH 047]
SUM_SOH allows to enable/disable each type of the failure events (and the relative detector) through the following configuration parameters:

Calibration Parameter	Detector
SOH_StatNvM_SupportEnabled	Status Failure/Integrity
SOH_ActNvM_SupportEnabled	Action Failure
SOH_PrivNvM_SupportEnabled	Privilege

If the detector is disabled, the relative detection counter and DID counters of the corresponding class are permanently set to the reserved value 0xFFFF.
[$Satisfies $SRS_SOH 103]

OS Errors Detector

[$SDD_SOH 069]

The SUM_SOH shall monitor the frequency of OS related errors and use the threshold during a time window filtering method to check for these occurrences. The SUM_SOH shall also implement a unique DID counter associated with each detection OS fault class

Hardware Protection

- Counter : SUM_SOH_ExcepOS_Cntr
- Counter Threshold : SUM_SOH_ExcepOS_Thresh
- Time Window : SUM_SOH_ExcepOS_TW

- Time Window Threshold : SUM_SOH_ExcepOS_TW_Thresh
- DID : SUM_SOH_DID_ExcepOS_Cntr

Service Protection and Trusted Function

- Counter : SUM_SOH_ServOS_Cntr
- Counter Threshold : SUM_SOH_ServOS_Thresh
- Time Window : SUM_SOH_ServOS_TW
- Time Window Threshold : SUM_SOH_ServOS_TW_Thresh
- DID : SUM_SOH_DID_ServOS_Cntr

Stack Fault

- Counter : SUM_SOH_StackOS_Cntr
- Counter Threshold : SUM_SOH_StackOS_Thresh
- Time Window : SUM_SOH_StackOS_TW
- Time Window Threshold : SUM_SOH_StackOS_TW_Thresh
- DID : SUM_SOH_DID_StackOS_Cntr

Memory Protection

- Counter : SUM_SOH_MemProtOS_Cntr
- Counter Threshold : SUM_SOH_MemProtOS_Thresh
- Time Window : SUM_SOH_MemProtOS_TW
- Time Window Threshold : SUM_SOH_MemProtOS_TW_Thresh
- DID : SUM_SOH_DID_MemProtOS_Cntr

Time Protection

- Counter : SUM_SOH_TimeProtOS_Cntr
- Counter Threshold : SUM_SOH_TimeProtOS_Thresh
- Time Window : SUM_SOH_TimeProtOS_TW
- Time Window Threshold : SUM_SOH_TimeProtOS_TW_Thresh
- DID : SUM_SOH_DID_TimeProtOS_Cntr

Other OS Errors

- Counter : SUM_SOH_ErrOS_Cntr
- Counter Threshold : SUM_SOH_ErrOS_Thresh
- Time Window : SUM_SOH_ErrOS_TW
- Time Window Threshold : SUM_SOH_ErrOS_TW_Thresh
- DID : SUM_SOH_DID_ErrOS_Cntr

[$Satisfies $SRS_SOH 104] [$Satisfies $SRS_SOH 105] [$Satisfies $SRS_SOH 106] [$Satisfies $SRS_SOH 108] [$Satisfies $SRS_SOH 109] [$Satisfies $SRS_SOH 110] [$Satisfies $SRS_SOH 111] [$Satisfies $SRS_SOH 112] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 073]
Callback called by the OS in its ProtectionHook and ErrorHook to identify the OS Error type and update the proper event detector (counter + filter).
[$Satisfies $SRS_SOH 114] [$Satisfies $SRS_SOH 115]

[$SDD_SOH 067]
The OS Errors shall be provided by ErrorHook and ProtectionHook Callbacks that are called in turn by OS when one of its API fails.

The **`SOH_OS_Protection_ErrorHook_Callback`** is called by ErrorHook and ProtectionHook to report OS Error type to SUM_SOH.

The mapping between OS Errors and detectors is the following :

Fault	Detector
E_OS_PROTECTION_EXCEPTION	Hardware Protection
E_OS_SERVICEID	Service Protection / Trusted Function
E_OS_ILLEGAL_ADDRESS	Service Protection / Trusted Function
E_OS_MISSINGEND	Service Protection / Trusted Function
E_OS_DISABLEDINT	Service Protection / Trusted Function
E_OS_STACKFAULT	Stack Fault
E_OS_PROTECTION_MEMORY	Memory Protection
E_OS_PROTECTION_TIME	Time Protection
E_OS_PROTECTION_ARRIVAL	Time Protection
E_OS_PROTECTION_LOCKED	Time Protection
Others	Other OS Errors

[$Satisfies $SRS_SOH 107] [$Satisfies $SRS_SOH 113] [$Satisfies $SRS_SOH 114]

[$SDD_SOH 068]
SUM_SOH allows to enable/disable each type of the OS failure (and the relative detector) through the following configuration parameters:

Calibration Parameter	Detector
SOH_ExcepOS_SupportEnabled	Hardware Protection
SOH_ServOS_SupportEnabled	Service Protection / Trusted Function
SOH_StackOS_SupportEnabled	Stack Fault
SOH_MemProtOS_SupportEnabled	Memory Protection
SOH_TimeProtOS_SupportEnabled	Time Protection
SOH_ErrOS_SupportEnabled	Other OS Errors

If the detector is disabled, the relative detection counter and DID counters of the corresponding class are permanently set to the reserved value 0xFFFF.
[$Satisfies $SRS_SOH 115]

3.4.3 Communication Reliability Monitor

[$SDD_SOH 059]
The scope of this monitor is to keep track of the following communication-related events:

- Clear Diagnostic Information Request
- Diagnostic Session Control Request
- SecurityAccess (Service 0x27) Failures
- MAC Failures
- SecP Errors.

[$Satisfies $SRS_SOH 133]

Clear Diagnostic Information Request

[$SDD_SOH 060]
***Clear Diagnostic Information Accepted Request* Detector**
To handle the *Clear Diagnostic Information Accepted Request* a Threshold filter is used with the following variables:

- Counter : SUM_SOH_ClearDiagInfo_Cntr
- Counter Threshold : SUM_SOH_ClearDiagInfo_Thresh
- DID : SUM_SOH_DID_ClearDiagInfo_Cntr

[$Satisfies $SRS_SOH 134] [$Satisfies $SRS_SOH 136] [$Satisfies $SRS_SOH 137] [$Satisfies $SRS_SOH 138] [$Satisfies $SRS_SOH 139] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 061]

***Clear Diagnostic Information Rejected Request* Detector**

To handle the *Clear Diagnostic Information Rejected Request* a Threshold filter is used with the following variables:

- Counter : SUM_SOH_Reject_ClearDiag_Cntr
- Counter Threshold : SUM_SOH_Reject_ClearDiag_Thresh
- DID : SUM_SOH_DID_Reject_ClearDiag_Cntr

[$Satisfies $SRS_SOH 140] [$Satisfies $SRS_SOH 142] [$Satisfies $SRS_SOH 143] [$Satisfies $SRS_SOH 144] [$Satisfies $SRS_SOH 145] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 062]

Both Clear Diagnostic Information Request events are provided by SUM_SSFM.
The SSFM writes in SSFM_ClearDiagReq to notify the SOH about a Clear Diagnostic Information Request and triggers `Runnable_SOH_CommunicationReliabilityMonitor_ClearDiag()`.
The ERRH can write the following values:

- 0x00 : Default
- 0x01 : Clear Diagnostic Request Approved, the SOH increases by one SUM_SOH_ClearDiagInfo_Cntr and calls `SOH_Filter()` for the *Clear Diagnostic Information Accepted Request* Detector
- 0x02 : Clear Diagnostic Request Rejected, the SOH increases by one SUM_SOH_Reject_ClearDiag_Cntr and calls `SOH_Filter()` for the *Clear Diagnostic Information Rejected Request* Detector
- 0x03-0x0F : Invalid.

[$Satisfies $SRS_SOH 135] [$Satisfies $SRS_SOH 141] [$Satisfies $SRS_SOH 239]

Diagnostic Session Control Request

[$SDD_SOH 063]

***Diagnostic Session Control Request* Detector**

To handle the *Diagnostic Session Control Request* a Threshold filter is used with the following variables:

- Counter : SUM_SOH_DiagSession_Cntr
- Counter Threshold : SUM_SOH_DiagSession_Thresh
- DID : SUM_SOH_DID_DiagSession_Cntr

[$Satisfies $SRS_SOH 146] [$Satisfies $SRS_SOH 147] [$Satisfies $SRS_SOH 148] [$Satisfies $SRS_SOH 149] [$Satisfies $SRS_SOH 150] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227] [$Satisfies $SRS_SOH 247]

The DIAG writes in DIAG_SOH_DiagSessionRequest to notify the SOH about a Diagnostic Session Request (service 0x10 subfunction 0x02) and triggers `Runnable_SOH_CommunicationReliabilityMonitor_DiagSession()`.
The DIAG can write the following values:

- FALSE : No Diagnostic Session Request (initial value)
- TRUE : Diagnostic Session Requested, the SOH increases by one SUM_SOH_DiagSession_Cntr and

calls `SOH_Filter()` for the *Diagnostic Session Control Request* Detector.

[$SDD_SOH 064]

***Accepted Diagnostic Session Control Request* Detector**

To handle the *Accepted Diagnostic Session Control Request* a Threshold filter is used with the following variables:

- Counter : SUM_SOH_Accept_DiagSession_Cntr
- Counter Threshold : SUM_SOH_Accept_DiagSession_Thresh
- DID : SUM_SOH_DID_Accept_DiagSession_Cntr

[$Satisfies $SRS_SOH 151] [$Satisfies $SRS_SOH 152] [$Satisfies $SRS_SOH 153] [$Satisfies $SRS_SOH 154] [$Satisfies $SRS_SOH 155] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

The Accept Diagnostic Session Control Request event is provided by BswM.

The BswM writes in the mode port DiagnosticReprogrammingRequest to notify the SOH about an Accepted Diagnostic Session Request and triggers `Runnable_SOH_CommunicationReliabilityMonitor_AcceptDiagSession()`.
The BswM can write the following modes:

- 0x00 : SUM_REPREQ_ACTIVE, the SOH increases by one SUM_SOH_Accept_DiagSession_Cntr and calls `SOH_Filter()` for the *Accepted Diagnostic Session Control Request* Detector
- 0x01 : SUM_REPREQ_INACTIVE (initial value).

SecurityAccess (Service 0x27) Failures

[$SDD_SOH 065]

***SecurityAccess (Service 0x27) Failures* Detector**

To handle the *SecurityAccess (Service 0x27) Failures* a Threshold filter is used with the following variables:

- Counter : SUM_SOH_SecurityAccess_Cntr
- Counter Threshold : SUM_SOH_SecurityAccess_Thresh
- DID : SUM_SOH_DID_SecurityAccess_Cntr

[$Satisfies $SRS_SOH 156] [$Satisfies $SRS_SOH 157] [$Satisfies $SRS_SOH 158] [$Satisfies $SRS_SOH 159] [$Satisfies $SRS_SOH 160] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227] [$Satisfies $SRS_SOH 245]

The SecurityAccess (Service 0x27) Failures events are provided by SSC.

The SSC writes in SSC_SOH_SecurityAccessError to notify the SOH about a SecurityAccess CompareKey (servcie 0x27) Failure and triggers `Runnable_SOH_CommunicationReliabilityMonitor_SecurityAccess()`.
The SSC can write the following values:

- FALSE : No SecurityAccess failures (initial value)
- TRUE : A SecurityAccess failure, the SOH increases by one SUM_SOH_SecurityAccess_Cntr and calls `SOH_Filter()` for the *SecurityAccess (Service 0x27) Failures* Detector.

MAC Failures

[$SDD_SOH 066]

***MAC Failures Types 1 and 2* Detector**

To handle the *MAC Failures Types 1 and 2* a Time Window filter is used with the following variables:

- Counter : SUM_SOH_MACfail_TenAct_Cntr
- Counter Threshold : SUM_SOH_MACfail_TenAct_Thresh
- Time Window : SUM_SOH_MACfail_TenAct_TW
- Time Window Threshold : SUM_SOH_MACfail_TenAct_TW_Thresh
- DID : SUM_SOH_DID_MACfail_TenAct_Cntr

[$Satisfies $SRS_SOH 166] [$Satisfies $SRS_SOH 167] [$Satisfies $SRS_SOH 168] [$Satisfies $SRS_SOH 169] [$Satisfies $SRS_SOH 170] [$Satisfies $SRS_SOH 171] [$Satisfies $SRS_SOH 172] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 070]

***MAC Failures Type 3* Detector**

To handle the *MAC Failures Type 3* a Time Window filter is used with the following variables:

- Counter : SUM_SOH_ReplayCount_Cntr
- Counter Threshold : SUM_SOH_ReplayCount_Thresh
- Time Window : SUM_SOH_ReplayCount_TW
- Time Window Threshold : SUM_SOH_ReplayCount_TW_Thresh
- DID : SUM_SOH_DID_ReplayCount_Cntr

[$Satisfies $SRS_SOH 173] [$Satisfies $SRS_SOH 174] [$Satisfies $SRS_SOH 175] [$Satisfies $SRS_SOH 176] [$Satisfies $SRS_SOH 177] [$Satisfies $SRS_SOH 178] [$Satisfies $SRS_SOH 179] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 071]

Respect to the SOH initialization, *MAC Failures Type 3* logs are delayed a number of seconds defined in the 1-byte calibration value SUM_SOH_ReplayCount_Start_Thresh.
This is a startup delay that allows the MAC counters to establish an initial synchronization before the detectors start counting errors.
Until the delay does not expire, the value of the boolean flag SUM_SOH_ReplayCount_Start_Status is FALSE.
`Runnable_SOH_CommunicationReliabilityMonitor_MacFailType3Delay()` runs periodically to check the elapsing time and - if the delay expired - flags SUM_SOH_ReplayCount_Start_Status to TRUE.
[$Satisfies $SRS_SOH 180] [$Satisfies $SRS_SOH 181] [$Satisfies $SRS_SOH 182]

[$SDD_SOH 072]
The three types of MAC Failures events are provided by SSC.
The SSC writes in SSC_E_SecureMessageMonitoringEvents to notify the SOH about a MAC Failure not related to SecP failures and triggers `Runnable_SOH_CommunicationReliabilityMonitor_MacFailType123()`.
The SSC can write the following values:

- 0x00 : SSC_ERR_NO_Event (initial value)
- 0x01 : SSC_ERR_MAC_Active, if the MACVerify failure event occurs when using the Active Counter while the tentative Counter state was invalid upon receipt of an authenticated message. In this case, SUM_SOH increases by one SUM_SOH_MACfail_TenAct_Cntr and calls `SOH_Filter()` for the *MAC Failures Types 1 and 2* Detector
- 0x02 : SSC_ERR_Tentative_Active, if the MACVerify failure event occurs while using the Active Counter after failed MAC verification using the Tentative Counter. In this case, SUM_SOH increases by one SUM_SOH_MACfail_TenAct_Cntr and calls `SOH_Filter()` for the *MAC Failures Types 1 and 2* Detector
- 0x03 : SSC_ERR_Low_Tentative, if the 64-bit counter received in a Counter Sync Message is not accepted by the receiver due to its value being less than or equal to the Active Counter. In this case, if SUM_SOH_ReplayCount_Start_Status is TRUE, SUM_SOH increases by one SUM_SOH_ReplayCount_Cntr and calls `SOH_Filter()` for the *MAC Failures Type 3* Detector.

[$Satisfies $SRS_SOH 161] [$Satisfies $SRS_SOH 162] [$Satisfies $SRS_SOH 163] [$Satisfies $SRS_SOH 164] [$Satisfies $SRS_SOH 165] [$Satisfies $SRS_SOH 182] [$Satisfies $SRS_SOH 244]

SecP Errors

[$SDD_SOH 052]
***SecP Errors* Detector**
To handle the *SecP Errors* a Time Window filter is used with the following variables:

- Counter : SUM_SOH_SecP_Cntr
- Counter Threshold : SUM_SOH_SecP_Thresh
- Time Window : SUM_SOH_SecP_TW
- Time Window Threshold : SUM_SOH_SecP_TW_Thresh
- DID : SUM_SOH_DID_SecP_Cntr

[$Satisfies $SRS_SOH 183] [$Satisfies $SRS_SOH 184] [$Satisfies $SRS_SOH 186] [$Satisfies $SRS_SOH 187] [$Satisfies $SRS_SOH 188] [$Satisfies $SRS_SOH 189] [$Satisfies $SRS_SOH 190] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 054]
The SOH component includes the RPort RP_ErrorStatus for receiving the HSM error notification from BswM. This Mode User Port refers to the Mode Declaration Group SECP_Error_Status and SOH reports the HSM error codes using the following mapping with SECP_Error_Status modes:

HSM Error Code	Enumeration	Mode Declaration
ERC_BUSY	0x02	SUM_ERC_BUSY
ERC_GENERAL_FAILURE	0xFF	SUM_ERC_GENERAL_FAILURE
ERC_KEY_EMPTY	0x12	SUM_ERC_KEY_EMPTY
ERC_KEY_INVALID	0x11	SUM_ERC_KEY_INVALID
ERC_KEY_NOT_AVAILABLE	0x10	SUM_ERC_KEY_NOT_AVAILABLE
ERC_KEY_UPDATE_ERROR	0x31	SUM_ERC_KEY_UPDATE_ERROR
ERC_KEY_WRITE_PROTECTED	0x30	SUM_ERC_KEY_WRITE_PROTECTED
ERC_MEMORY_FAILURE	0x50	SUM_ERC_MEMORY_FAILURE
ERC_NO_DEBUGGING	0x60	SUM_ERC_NO_DEBUGGING
ERC_NO_ERROR	0x00	SUM_ERC_NO_ERROR
ERC_NO_SECURE_BOOT	0x20	SUM_ERC_NO_SECURE_BOOT
ERC_SEQUENCE_ERROR	0x01	SUM_ERC_SEQUENCE_ERROR
ERC_RNG_SEED	0x40	SUM_ERC_RNG_SEED

For the correct error code reporting SOH assumes that BswM configuration includes the definition of:

- a Mode Switch Port to be connected to RP_ErrorStatus;
- a Mode Request Port to receive mode change requests from Cry module;
- a Rule for each mode change notification to SOH.

For all the values different than 0x00, the SOH increases by one SUM_SOH_SecP_Cntr and calls `SOH_Filter()` for the *SecP Errors* Detector.
[$Satisfies $SRS_SOH 185] [$Satisfies $SRS_SOH 251]

3.4.4 Watchdog Manager Reliability Monitor

[$SDD_SOH 094]
The scope of this monitor is to track the occurrences of: Failed Local Supervisions of Supervised Entities (except ERRH and SOH), WDGM_E_SUPERVISION or WDGM_E_SET_MODE errors, and SUM_ERRH Failed Local Supervisions.
[$Satisfies $SRS_SOH 191] [$Satisfies $SRS_SOH 192] [$Satisfies $SRS_SOH 193] [$Satisfies $SRS_SOH 200] [$Satisfies $SRS_SOH 216]

[$SDD_SOH 095]

Local Supervision Failures Detector

To handle the Failed Local Supervision, a Time Window filter is used with the following variables :

- Counter : SUM_SOH_LocalStatFail_Cntr
- Counter Threshold : SUM_SOH_LocalStatFail_Thresh
- Time Window : SUM_SOH_LocalStatFail_TW
- Time Window Threshold : SUM_SOH_LocalStatFail_TW_Thresh
- DID : SUM_SOH_DID_LocalStatFail_Cntr

[$Satisfies $SRS_SOH 194] [$Satisfies $SRS_SOH 195] [$Satisfies $SRS_SOH 196] [$Satisfies $SRS_SOH 197] [$Satisfies $SRS_SOH 198] [$Satisfies $SRS_SOH 199] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 098]

The WdgM provides the events of Failed Local Supervisions of Supervised Entities through the corresponding WdgM_LocalMode interface. In particular, for each Supervised Entity, the WdgM generates a port whose interface is WdgM_LocalMode.
For each supervised entity (except ERRH and SOH), there exists a **`Runnable_WdgMReliabilityMonitor_FailedLocalSupervision_<Supervised_Entity_name>`** which is invoked when one of the following three events occurs:

1. The local supervision status (read at the corresponding SOH_WdgM_IndividualMode port) of the supervised entity transitions from SUPERVISION_OK (0) to SUPERVISION_FAILED (1).
2. The local supervision status (read at the corresponding SOH_WdgM_IndividualMode port) of the supervised entity transitions from SUPERVISION_OK (0) to SUPERVISION_EXPIRED (2).
3. The local supervision status (read at the corresponding SOH_WdgM_IndividualMode port) of the supervised entity transitions from SUPERVISION_FAILED (1) to SUPERVISION_EXPIRED (2).

These runnables increment the counter `SUM_SOH_LocalStatFail_Cntr` that they share. They also call the `SOH_Filter` and if the `SUM_SOH_DID_LocalStatFail_Cntr` is changed and the battery voltage is in range, the `SUM_SOH_DID_LocalStatFail_Cntr` is saved into the NvM.
The SOH guarantees a synchronized access to `SUM_SOH_LocalStatFail_Cntr` and the other shared variables (that the runnables access in the `SOH_Filter` function) by defining proper `EXCLUSIVE-AREAS`.

[$Satisfies $SRS_SOH 027] [$Satisfies $SRS_SOH 265] [$Satisfies $SRS_SOH 029] [$Satisfies $SRS_SOH 038] [$Satisfies $SRS_SOH 047] [$Satisfies $SRS_SOH 192] [$Satisfies $SRS_SOH 193] [$Satisfies $SRS_SOH 194] [$Satisfies $SRS_SOH 249] [$Satisfies $SRS_SOH 250]

[$SDD_SOH 096]

WdgM DEM Supervision Errors Detector

To handle the WDGM_E_SUPERVISION and WDGM_E_SET_MODE errors, a Time Window filter is used with the following variables :

- Counter : SUM_SOH_WdgM_DEMErr_Cntr
- Counter Threshold : SUM_SOH_WdgM_DEMErr_Thresh
- Time Window : SUM_SOH_WdgM_DEMErr_TW
- Time Window Threshold : SUM_SOH_WdgM_DEMErr_TW_Thresh
- DID : SUM_SOH_DID_WdgM_DEMErr_Cntr

[$Satisfies $SRS_SOH 200] [$Satisfies $SRS_SOH 201] [$Satisfies $SRS_SOH 202] [$Satisfies $SRS_SOH 203] [$Satisfies $SRS_SOH 204] [$Satisfies $SRS_SOH 205] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 099]

The events of WDGM_E_SUPERVISION and WDGM_E_SET_MODE are provided by the ERRH.
The **`Runnable_SOH_ERRH_BswEvent`** is invoked when the value at the ERRH_E_BswEvent interface changes.
If one of the following events occurs:

- WDGM_E_SUPERVISION
- WDGM_E_SET_MODE

the runnable increments the `SUM_SOH_WdgM_DEMErr_Cntr`. It also calls the `SOH_Filter` and if the `SUM_SOH_DID_WdgM_DEMErr_Cntr` is changed and the battery voltage is in range, the `SUM_SOH_DID_WdgM_DEMErr_Cntr` is saved into the NvM.

[$Satisfies $SRS_SOH 027] [$Satisfies $SRS_SOH 265] [$Satisfies $SRS_SOH 029] [$Satisfies $SRS_SOH 038] [$Satisfies $SRS_SOH 047] [$Satisfies $SRS_SOH 200] [$Satisfies $SRS_SOH 228] [$Satisfies $SRS_SOH 229] [$Satisfies $SRS_SOH 238]

[$SDD_SOH 097]

SUM_ERRH Local Supervision Detector

To handle the SUM_ERRH local supervision failures a Time Window filter is used with the following variables :

- Counter : SOH_ERRHSupFail_Cntr
- Counter Threshold : SOH_ERRHSupFail_Thresh
- Time Window : SOH_ERRHSupFail_TW
- Time Window Threshold : SOH_ERRHSupFail_TW_Thresh
- DID : SUM_SOH_DID_ERRHSupFail_Cntr

[$Satisfies $SRS_SOH 218] [$Satisfies $SRS_SOH 219] [$Satisfies $SRS_SOH 220] [$Satisfies $SRS_SOH 221] [$Satisfies $SRS_SOH 222] [$Satisfies $SRS_SOH 223] [$Satisfies $SRS_SOH 226] [$Satisfies $SRS_SOH 227]

[$SDD_SOH 100]
The WdgM provides the events of Failed Local Supervisions of the ERRH through the corresponding WdgM_LocalMode interface. The **`Runnable_SOH_WdgMReliabilityMonitor_ERRHFailedLocalSupervision`** is is invoked when one of the following three events occurs:

1. The local supervision status (read at the SOH_WdgM_IndividualMode_Errh interface) of the ERRH transitions from SUPERVISION_OK (0) to SUPERVISION_FAILED (1).
2. The local supervision status (read at the SOH_WdgM_IndividualMode_Errh interface) of the ERRH transitions from SUPERVISION_OK (0) to SUPERVISION_EXPIRED (2).
3. The local supervision status (read at the SOH_WdgM_IndividualMode_Errh interface) of the ERRH transitions from SUPERVISION_FAILED (1) to SUPERVISION_EXPIRED (2).

This runnable increments the `SOH_ERRHSupFail_Cntr` and calls the `SOH_Filter`. If the `SUM_SOH_DID_ERRHSupFail_Cntr` is changed and the battery voltage is in range, the `SUM_SOH_DID_ERRHSupFail_Cntr` is saved into the NvM. The `ERRH_Local_Supervision_Status` field of the DID 0xF093, will be instead updated directly in the DID function which, when called, reads the ERRH Local Supervision Status.

[$Satisfies $SRS_SOH 027] [$Satisfies $SRS_SOH 265] [$Satisfies $SRS_SOH 029] [$Satisfies $SRS_SOH 038] [$Satisfies $SRS_SOH 047] [$Satisfies $SRS_SOH 217] [$Satisfies $SRS_SOH 218] [$Satisfies $SRS_SOH 224] [$Satisfies $SRS_SOH 225] [$Satisfies $SRS_SOH 249]

[$SDD_SOH 101]
SUM_SOH Local Supervision Detector
SUM_ERRH implements this detector. However, to guarantee its proper behaviour the SUM_SOH executes the logic needed to update the corresponding WdgM_LocalSupervision port interface.
[$Satisfies $SRS_SOH 207][$Satisfies $SRS_SOH 262]

3.4.5 Platform Reliability Monitor

This feature is described in the Architecture Design document.

3.4.6 Data Provider

SUM_SOH provides data regarding monitored events via DIDs allocated for the SUM_SOH and SUM_ERRH.

3.5 RID 0x02C0 Clear SOH Logs

[$SDD_SOH 038]
Through the UDS Routine 0x02C0 (Clear SUM_SOH Logs) the tester can clear detectors, timers, DIDs and counters for all the SOH monitors. This routine implements only the SubFunction 0x1 (Start Routine) with 2 unsigned bytes as Option Record.
The accepted values for the Option Record are :

- 0xF08F : clear DID F08F metrics
- 0xF090 : clear DID F090 metrics
- 0xF091 : clear DID F091 metrics
- 0xF092 : clear DID F092 metrics
- 0xF093 : clear DID F093 metrics
- 0xF094 : asks the ERRH to clear DID F094 metrics
- 0xFFFF : clear all metrics

This RID must be created by the user in the DCM module so that after BSW generation the interface **RoutineServices_Clear_SUM_SOH_logs** is available.
[$Satisfies $SRS_SOH 040] [$Satisfies $SRS_SOH 069] [$Satisfies $SRS_SOH 266]

[$SDD_SOH 039]
SUM_SOH implements the **Runnable_SOH_RIDClearLogs** that is the Server part of the interface *RoutineServices_Clear_SUM_SOH_logs*. It's called through an Operation Invoked Event when the *Start Routine* for the Routine 0x02C0 (Clear SUM_SOH Logs) is invoked through the UDS protocol. It clears the needed metrics based on the Option Record received value.
The runnable behaviour is described in the diagram 3.2 .
[$Satisfies $SRS_SOH 031] [$Satisfies $SRS_SOH 043] [$Satisfies $SRS_SOH 044] [$Satisfies $SRS_SOH 056] [$Satisfies $SRS_SOH 060] [$Satisfies $SRS_SOH 061] [$Satisfies $SRS_SOH 069] [$Satisfies $SRS_SOH 266]

[$SDD_SOH 102]
When the DCM calls the RID with Option Record values 0xF094 or 0xFFFF, the SUM_SOH invokes a client operation service in the SUM_ERRH (Acting as a server) to clean all the metrics related to SUM_SOH Local Supervision Detector implemented at the ERRH side. To complete the RID 0x02C0 routine the SUM_SOH shall interact with the SUM_ERRH through the Client-Server interface SOH_ERRH_RID_Clear_Logs where SUM_ERRH acts as a Server.
[$Satisfies $SRS_SOH 260] [$Satisfies $SRS_SOH 261]

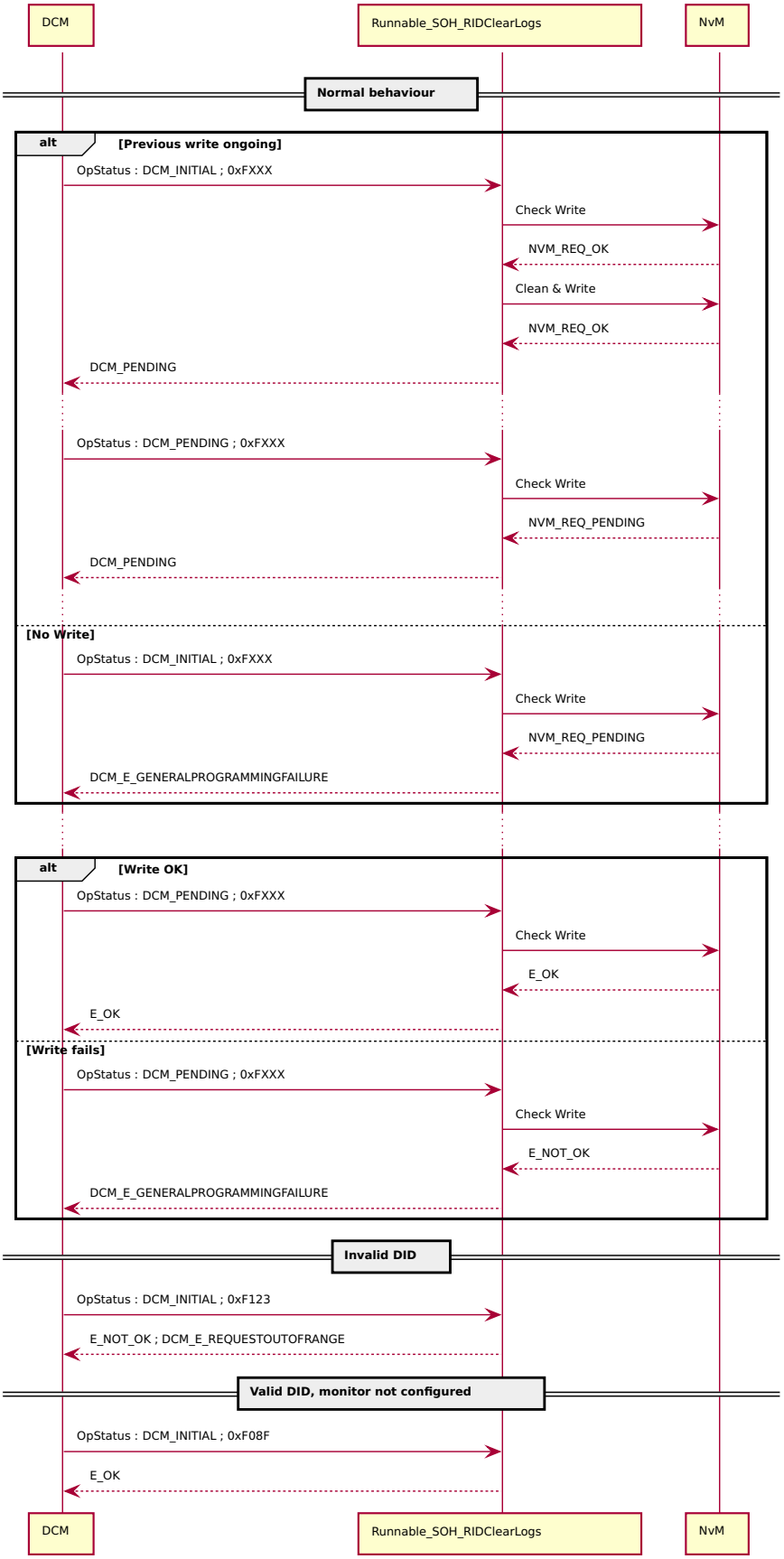


Diagram 3.2: Runnable_SOH_RIDClearLogs

3.6 DID Counters

3.6.1 DID 0xF08F: SUM_SOH Reset Reliability Monitor

[$SDD_SOH 031]
It shall export the results of reset monitoring events that occur in an ECU. They are the following 2-byte unsigned integer :

- SUM_SOH_DID_ImmeReset_Cntr
- SUM_SOH_DID_DelayedReset_Cntr
- SUM_SOH_DID_DiagReset_Cntr

[$Satisfies $SRS_SOH 034]

Through operation invoked events those values are passed to the DCM .Four Runnables are used :

[$SDD_SOH 040]
Runnable_SOH_DID_F08F_Data_ImmeReset_Cntr to pass SUM_SOH_DID_ImmeReset_Cntr.
[$Satisfies $SRS_SOH 034]

[$SDD_SOH 041]
Runnable_SOH_DID_F08F_Data_DiagReset_Cntr to pass SUM_SOH_DID_DiagReset_Cntr.
[$Satisfies $SRS_SOH 034]

[$SDD_SOH 042]
Runnable_SOH_DID_F08F_Data_DelayedReset_Cntr to pass SUM_SOH_DID_DelayedReset_Cntr.
[$Satisfies $SRS_SOH 034]

[$SDD_SOH 043]
Runnable_SOH_DID_F08F_CheckRead to allow the read.
[$Satisfies $SRS_SOH 034]

3.6.2 DID 0xF090: SUM_SOH Memory Status Reliability Monitor

[$SDD_SOH 032]
It shall maintain the results of monitoring NVM production errors and OS errors that occur in an ECU.

- SUM_SOH_DID_StatNVM_Cntr
- SUM_SOH_DID_ActNVM_Cntr
- SUM_SOH_DID_PrivNVM_Cntr
- SUM_SOH_DID_ExcepOS_Cntr
- SUM_SOH_DID_ServOS_Cntr
- SUM_SOH_DID_StackOS_Cntr
- SUM_SOH_DID_MemProtOS_Cntr
- SUM_SOH_DID_TimeProtOS_Cntr
- SUM_SOH_DID_ErrOS_Cntr

[$Satisfies $SRS_SOH 035] [$Satisfies $SRS_SOH 112] [$Satisfies $SRS_SOH 113] [$Satisfies $SRS_SOH 114]

Through operation invoked events those values are passed to the DCM .Ten Runnables are used :

[$SDD_SOH 048]
Runnable_SOH_DID_F090_Data_StatNVM_Cntr to pass SUM_SOH_DID_StatNVM_Cntr.
[$Satisfies $SRS_SOH 035]

[$SDD_SOH 049]
Runnable_SOH_DID_F090_Data_ActNVM_Cntr to pass SUM_SOH_DID_ActNVM_Cntr.
[$Satisfies $SRS_SOH 035]

[$SDD_SOH 050]
Runnable_SOH_DID_F090_Data_PrivNVM_Cntr to pass SUM_SOH_DID_PrivNVM_Cntr.
[$Satisfies $SRS_SOH 035]

[$SDD_SOH 051]
Runnable_SOH_DID_F090_Data_ExcepOS_Cntr to pass SUM_SOH_DID_ExcepOS_Cntr.
[$Satisfies $SRS_SOH 035]

[$SDD_SOH 083]
Runnable_SOH_DID_F090_Data_ServOS_Cntr to pass SUM_SOH_DID_ServOS_Cntr.
[$Satisfies $SRS_SOH 035]

[$SDD_SOH 053]
Runnable_SOH_DID_F090_Data_StackOS_Cntr to pass SUM_SOH_DID_StackOS_Cntr.
[$Satisfies $SRS_SOH 035]

[$SDD_SOH 085]
Runnable_SOH_DID_F090_Data_MemProtOS_Cntr to pass SUM_SOH_DID_MemProtOS_Cntr.
[$Satisfies $SRS_SOH 035]

[$SDD_SOH 055]
Runnable_SOH_DID_F090_Data_TimeProtOS_Cntr to pass SUM_SOH_DID_TimeProtOS_Cntr.
[$Satisfies $SRS_SOH 035]

[$SDD_SOH 056]
Runnable_SOH_DID_F090_Data_ErrOS_Cntr to pass SUM_SOH_DID_ErrOS_Cntr.
[$Satisfies $SRS_SOH 035]

[$SDD_SOH 084]
Runnable_SOH_DID_F090_CheckRead to allow the read.
[$Satisfies $SRS_SOH 035]

3.6.3 DID 0xF091: SUM_SOH Platform Reliability Monitor

This feature is described in the Architecture Design document.

3.6.4 DID 0xF092: SUM_SOH Communication Reliability Monitor

[$SDD_SOH 034]
It shall maintain the results of monitoring communication related events.

- SUM_SOH_DID_ClearDiagInfo_Cntr
- SUM_SOH_DID_Reject_ClearDiag_Cntr
- SUM_SOH_DID_DiagSession_Cntr
- SUM_SOH_DID_Accept_DiagSession_Cntr
- SUM_SOH_DID_SecurityAccess_Cntr
- SUM_SOH_DID_MACfail_TenAct_Cntr
- SUM_SOH_DID_ReplayCount_Cntr
- SUM_SOH_DID_SecP_Cntr

Each one of them is a 2-byte unsigned integer.
They are used to update the DCM DID 0xF092 by Runnable_SOH_DID_F092 through an operation invoked event.
[$Satisfies $SRS_SOH 037]

Through operation invoked events those values are passed to the DCM .Four Runnables are used :

[$SDD_SOH 074]
Runnable_SOH_DID_F092_Data_ClearDiagInfo_Cntr to pass SUM_SOH_DID_ClearDiagInfo_Cntr.
[$Satisfies $SRS_SOH 037]

[$SDD_SOH 075]
Runnable_SOH_DID_F092_Data_Reject_ClearDiag_Cntr to pass SUM_SOH_DID_Reject_ClearDiag_Cntr.
[$Satisfies $SRS_SOH 037]

[$SDD_SOH 076]
Runnable_SOH_DID_F092_Data_DiagSession_Cntr to pass SUM_SOH_DID_DiagSession_Cntr.
[$Satisfies $SRS_SOH 037]

[$SDD_SOH 077]
Runnable_SOH_DID_F092_Data_Accept_DiagSession_Cntr to pass SUM_SOH_DID_Accept_DiagSession_Cntr.

[$Satisfies $SRS_SOH 037]

[$SDD_SOH 078]
Runnable_SOH_DID_F092_Data_SecurityAccess_Cntr to pass SUM_SOH_DID_SecurityAccess_Cntr.
[$Satisfies $SRS_SOH 037]

[$SDD_SOH 079]
Runnable_SOH_DID_F092_Data_MACfail_TenAct_Cntr to pass SUM_SOH_DID_MACfail_TenAct_Cntr.
[$Satisfies $SRS_SOH 037]

[$SDD_SOH 080]
Runnable_SOH_DID_F092_Data_ReplayCount_Cntr to pass SUM_SOH_DID_ReplayCount_Cntr.
[$Satisfies $SRS_SOH 037]

[$SDD_SOH 081]
Runnable_SOH_DID_F092_Data_SecP_Cntr to pass SUM_SOH_DID_SecP_Cntr.
[$Satisfies $SRS_SOH 037]

[$SDD_SOH 082]
Runnable_SOH_DID_F092_CheckRead to allow the read.
[$Satisfies $SRS_SOH 034]

3.6.5 DID 0xF093: SUM_SOH Watchdog Manager Reliability Monitor

[$SDD_SOH 035]
It shall maintain the results of monitoring communication related events.

- SUM_SOH_DID_LocalStatFail_Cntr
- SUM_SOH_DID_WdgM_DEMErr_Cntr
- SUM_SOH_DID_ERRHSupFail_Cntr
- ERRH_Local_Supervision_Status

Each one of them is a 2-byte unsigned integer except ERRH_Local_Supervision_Status is 1-byte unsigned integer.

[$Satisfies $SRS_SOH 038]

Through operation invoked events those values are passed to the DCM .Five Runnables are used :

[$SDD_SOH 104]
Runnable_SOH_DID_F093_Data_LocalStatFail_Cntr to pass SUM_SOH_DID_LocalStatFail_Cntr
[$Satisfies $SRS_SOH 038]

[$SDD_SOH 105]
Runnable_SOH_DID_F093_Data_Data_WdgM_DEMErr_Cntr to pass SUM_SOH_DID_WdgM_DEMErr_Cntr
[$Satisfies $SRS_SOH 038]

[$SDD_SOH 106]
Runnable_SOH_DID_F093_Data_ERRHSupFail_Cntr to pass SUM_SOH_DID_ERRHSupFail_Cntr
[$Satisfies $SRS_SOH 038]

[$SDD_SOH 107]
Runnable_SOH_DID_F093_Data_ERRH_Local_Supervision_Status to pass ERRH_Local_Supervision_Status

[$Satisfies $SRS_SOH 038]

[$SDD_SOH 108]
Runnable_SOH_DID_F093_CheckRead to allow the read.
[$Satisfies $SRS_SOH 038]

3.7 Configuration

[$SDD_SOH 112]
SUM_SOH is a SWC within the context of AUTOSAR. It consists of static files and generated files. SUM_SOH configuration is done the same way as for standard BSW modules. A parameter definition file is included with the module which allows users to configure the available parameters. The generated files change based on the configuration.
For default values of calibration thresholds SOH is referring to the SDV-8002. Configuration parameters shall be pre-compile, in other words the ECU behavior modification requires a reflash of the complete application software.
[$Satisfies $SRS_SOH 070] [$Satisfies $SRS_SOH 254] [$Satisfies $SRS_SOH 255] [$Satisfies $SRS_SOH 256]

3.7.1 Included Containers

Name	Multiplicity	Scope / Dependency
SOHGeneral	1..1	General configuration parameters of the SOH
SUSDConfigSet	1..1	This container contains the configuration parameters and sub containers of the SOH module supporting multiple SWC configuration sets

3.7.2 SOHGeneral

Parameter Name	SOH_PermitExecution_Mask
Description	Mask that can be OR'd with the value supplied by PermitExecution to set any "don't care" bits to '1'.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..255
Default value	190

Parameter Name	SOH_RunnableSOHMainPeriod
Description	Time in sec, for the periodicity of the Runnable_SOH_Main.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0.001..0.2
Default value	0.1

[$SDD_SOH 036]

Parameter Name	SohGenerateInitRunnable
Description	This parameter, when configured to TRUE, generates a periodic Runnable Entity that is useful for calling the SOH_Init() function just once.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	0

Parameter Name	SohInitFuncPeriodicity
Description	This parameter is used to specify the time in ms for the periodicity of the runnable Runnable_SOH_Init.
Multiplicity (upper..lower)	0..1
Category	ECUC-FLOAT-PARAM-DEF
Range	1.0...5000.0
Default value	1000.0

[$Satisfies $SRS_SOH 259]

Parameter Name	SOH_StatNvM_SupportEnabled
Description	This parameter is used to enable the Nvm Status Failure/Integrity Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

Parameter Name	SOH_ActNvM_SupportEnabled
Description	This parameter is used to enable the Nvm Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

Parameter Name	SOH_PrivNvM_SupportEnabled
Description	This parameter is used to enable the Nvm Privilege class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

Parameter Name	SOH_ExcepOS_SupportEnabled
Description	This parameter is used to enable the OS Hardware Protection Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_ServOS_SupportEnabled
Description	This parameter is used to enable the OS Service Protection / Trusted Function Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_StackOS_SupportEnabled
Description	This parameter is used to enable the OS Stack Fault class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_MemProtOS_SupportEnabled
Description	This parameter is used to enable the OS Memory Protection Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_TimeProtOS_SupportEnabled
Description	This parameter is used to enable the OS Time Protection Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_ErrOS_SupportEnabled
Description	This parameter is used to enable the OS Other OS Errors Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_RID_02C0_SOH_Clear_Log_Reference
Description	Reference to DCM RID 0x02C0 SOH Clear Logs
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_ImmediateDelayed_SupportEnabled
Description	It allows to enable (disable) the Immediate Reset Detector and Delayed Reset Detector in Reset Reliability monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_DiagnosticReset_SupportEnabled
Description	It allows to enable (disable) the Diagnostic Reset Detector in Reset Reliability monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_ClearDiagnosticInformation_SupportEnabled
Description	It allows to enable (disable) the Clear Diagnostic Information Accepted Request Detector and Clear Diagnostic Information Rejected Request Detector in Communication Reliability Monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_DiagnosticSessionControl_SupportEnabled
Description	It allows to enable (disable) the Diagnostic Session Control Request Detector and Accepted Diagnostic Session Control Request Detector in Communication Reliability Monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_SecurityAccessFailures_SupportEnabled
Description	It allows to enable (disable) the SecurityAccess (Service 0x27) Failures Detector in Communication Reliability Monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_MACFailures_SupportEnabled
Description	It allows to enable (disable) the MAC Failures Types 1 and 2 Detector and MAC Failures Type 3 Detector in Communication Reliability Monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_SecPErrors_SupportEnabled
Description	It allows to enable (disable) the SecP Errors Detector in Communication Reliability Monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_LocalSupervisionFailures_SupportEnabled
Description	It allows to enable (disable) the Local Supervision Failures Detector in Watchdog Manager Reliability Monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_WdgMDEMSupervisionErrors_SupportEnabled
Description	It allows to enable (disable) the WdgM DEM Supervision Errors Detector in Watchdog Manager Reliability Monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_ERRHLocalSupervision_SupportEnabled
Description	It allows to enable (disable) the SUM_ERRH Local Supervision Detector in Watchdog Manager Reliability Monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	ERRH_SOHLocalSupervision_SupportEnabled
Description	It allows to enable (disable) the SUM_SOH Local Supervision Detector implemented by the ERRH.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

Parameter Name	SOH_ShutdownDelay_SupportEnabled
Description	It allows to enable (disable) the Shutdown Delay Detector in Platform Reliability Monitor.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	TRUE

3.7.3 SOHConfigSet

SOH_ResetReliabilityMonitor

This container includes the configuration parameters needed for the Reset Reliability Monitor detectors.

SOH_ImmediateReset

Parameter Name	SUM_SOH_ImmeReset_Thresh_Init
Description	Initial value of SUM_SOH_ImmeReset_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ImmeReset_TW_Thresh_Init
Description	Initial value of SUM_SOH_ImmeReset_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMImmResetDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F08F_ImmResetData_Reference
Description	Reference to DCM ImmReset Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_DelayedReset

Parameter Name	SUM_SOH_DelayedReset_Thresh_Init
Description	Initial value of SUM_SOH_DelayedReset_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_DelayedReset_TW_Thresh_Init
Description	Initial value of SUM_SOH_DelayedReset_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMDelayedResetDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F08F_DelayedResetData_Reference
Description	Reference to DCM DelayedReset Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_DiagnosticReset

Parameter Name	SUM_SOH_DiagReset_Thresh_Init
Description	Initial value SUM_SOH_DiagReset_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMDiagResetCntr_Reference
Description	Reference to the NVM Counter Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMDiagResetDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F08F_DiagResetData_Reference
Description	Reference to DCM DiagReset Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_MemoryStatusReliabilityMonitor

This container includes the configuration parameters needed for the Memory Status Monitor detectors.

SOH_NVMErrorsDetector

This container includes the configuration containers needed for the NVM Error detectors.

SOH_StatusNVM

Parameter Name	SUM_SOH_StatNVM_Thresh_Init
Description	Initial value of SUM_SOH_StatNVM_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_StatNVM_TW_Thresh_Init
Description	Initial value of SUM_SOH_StatNVM_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMStatNVMDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_StatusNVM_Reference
Description	Reference to DCM StatusNVM Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_ActNVM

Parameter Name	SUM_SOH_ActNVM_Thresh_Init
Description	Initial value of SUM_SOH_ActNVM_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ActNVM_TW_Thresh_Init
Description	Initial value of SUM_SOH_ActNVM_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMActNVMDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_ActNVM_Reference
Description	Reference to DCM ActNVM Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_PrivNVM

Parameter Name	SUM_SOH_PrivNVM_Thresh_Init
Description	Initial value of SUM_SOH_PrivNVM_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_PrivNVM_TW_Thresh_Init
Description	Initial value of SUM_SOH_PrivNVM_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMPrivNVMDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_PrivNVM_Reference
Description	Reference to DCM PrivNVM Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_OSErrorsDetector
The following configuration parameters are used to set the initial threshold value for each OS error class detection counter and their related time windows.

SOH_ExcepOS

Parameter Name	SUM_SOH_ExcepOS_Thresh_Init
Description	Initial value of SUM_SOH_ExcepOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ExcepOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_ExcepOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMExcepOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_ExcepOS_Reference
Description	Reference to DCM ExcepOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_ServOS

Parameter Name	SUM_SOH_ServOS_Thresh_Init
Description	Initial value of SUM_SOH_ServOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ServOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_ServOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMServOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_ServOS_Reference
Description	Reference to DCM ServOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_StackOS

Parameter Name	SUM_SOH_StackOS_Thresh_Init
Description	Initial value of SUM_SOH_StackOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_StackOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_StackOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMStackOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_StackOS_Reference
Description	Reference to DCM StackOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_MemProtOS

Parameter Name	SUM_SOH_MemProtOS_Thresh_Init
Description	Initial value of SUM_SOH_MemProtOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_MemProtOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_MemProtOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMMemProtOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_MemProtOS_Reference
Description	Reference to DCM MemProtOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_TimeProtOS

Parameter Name	SUM_SOH_TimeProtOS_Thresh_Init
Description	Initial value of SUM_SOH_TimeProtOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_TimeProtOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_TimeProtOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMTimeProtOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_TimeProtOS_Reference
Description	Reference to DCM TimeProtOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_ErrOS

Parameter Name	SUM_SOH_ErrOS_Thresh_Init
Description	Initial value of SUM_SOH_ErrOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ErrOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_ErrOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMErrOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_ErrOS_Reference
Description	Reference to DCM ErrOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SUM_SOH_ShutDown_Thresh_Init
Description	Initial value of SUM_SOH_ShutDown_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ShutDown_Timer_Thresh_Init
Description	Initial value of SUM_SOH_ShutDown_Timer_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	60

SOH_CommunicationReliabilityMonitor

This container includes the configuration parameters needed for the Communication Reliability Monitor detectors. Here organized by events.

SOH_ClearDiagInfo

Parameter Name	SUM_SOH_ClearDiagInfo_Thresh_Init
Description	Initial value of SUM_SOH_ClearDiagInfo_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMClearDiagInfoDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_ClearDiagInfoDID_Reference
Description	Reference to DCM Clear Diag Infomation Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_RejectClearDiag

Parameter Name	SUM_SOH_Reject_ClearDiag_Thresh_Init
Description	Initial value of SUM_SOH_Reject_ClearDiag_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMRejectClearDiagDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_RejectClearDiagDID_Reference
Description	Reference to DCM Reject Clear Diag Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_DiagSession

Parameter Name	SUM_SOH_DiagSession_Thresh_Init
Description	Initial value of SUM_SOH_DiagSession_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMDiagSessionCntr_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMDiagSessionDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_DiagSessionDID_Reference
Description	Reference to DCM Diag Session Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_Accept_DiagSession

Parameter Name	SUM_SOH_Accept_DiagSession_Thresh_Init
Description	Initial value of SUM_SOH_Accept_DiagSession_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMAccept_DiagSessionCntr_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMAccept_DiagSessionDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_Accept_DiagSessionDID_Reference
Description	Reference to DCM Accept Diag Session Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_SecurityAccess

Parameter Name	SUM_SOH_SecurityAccess_Thresh_Init
Description	Initial value of SUM_SOH_SecurityAccess_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMSecurityAccessCntr_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMSecurityAccessDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_SecurityAccessDID_Reference
Description	Reference to DCM Security Access Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_MACfailTenAct

Parameter Name	SUM_SOH_MACfail_TenAct_Thresh_Init
Description	Initial value of SUM_SOH_MACfail_TenAct_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	$2\times$ number of received MAC protected messages

Parameter Name	SUM_SOH_MACfail_TenAct_TW_Thresh_Init
Description	Initial value of SUM_SOH_MACfail_TenAct_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	2

Parameter Name	SOH_NvMMACfail_TenActDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_MACfail_TenActDID_Reference
Description	Reference to DCM MAC fail type 1 and 2 Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_ReplayCount

Parameter Name	SUM_SOH_ReplayCount_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ReplayCount_TW_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SUM_SOH_ReplayCount_Start_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_Start_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	5

Parameter Name	SOH_NvMReplayCountDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_ReplayCountDID_Reference
Description	Reference to DCM MAC fail type 3 Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_SecP

Parameter Name	SUM_SOH_SecP_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_SecP_TW_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMSecPDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_SecPDID_Reference
Description	Reference to DCM SecP Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_WatchdogManagerReliabilityMonitor

This container includes the configuration parameters needed for the Watchdog Manager Reliability Monitor detectors.

Local_Supervision_Failures_Detector

Parameter Name	SUM_SOH_ImmeReset_Thresh_Init
Description	Initial value of SUM_SOH_LocalStatFail_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_LocalStatFail_TW_Thresh_Init
Description	Initial value of SUM_SOH_LocalStatFail_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	10

Parameter Name	Supervised_Entity_Reference
Description	Reference to a Supervised Entity container in the WdgM
Multiplicity (upper..lower)	0..Infinite
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_LocalSupervisionFailures_DID_counter_Reference
Description	Reference to the NVM Block related to SUM_SOH_DID_LocalStatFail_Cntr
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F093_LocalStatFail_Reference
Description	Reference to DCM LocalStatFail Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

WdgM_DEM_Supervision_Errors_Detector

Parameter Name	SUM_SOH_WdgM_DEMErr_Thresh_Init
Description	Initial value of SUM_SOH_WdgM_DEMErr_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_WdgM_DEMErr_TW_Thresh_Init
Description	Initial value of SUM_SOH_WdgM_DEMErr_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	10

Parameter Name	SOH_WdgMDEMErr_DID_counter_Reference
Description	Reference to the NVM Block related to SUM_SOH_DID_WdgM_DEMErr_Cntr
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F093_WdgMDEMErr_Reference
Description	Reference to DCM WdgMDEMErr Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SUM_ERRH_Local_Supervision_Detector

Parameter Name	SOH_ERRHSupFail_Thresh_Init
Description	Initial value of SOH_ERRHSupFail_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_ERRHSupFail_TW_Thresh_Init
Description	Initial value of SOH_ERRHSupFail_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	10

Parameter Name	SOH_ERRHLocalSupervisionFailures_DID_counter_Reference
Description	Reference to the NVM Block related to SUM_SOH_DID_ERRHSupFail_Cntr
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F093_ERRHSupFail_Reference
Description	Reference to DCM ERRHSupFail Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F093_ERRH_Local_Supervision_Status_Reference
Description	Reference to DCM ERRH Local Supervision Status Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_PlatformReliabilityMonitor

This container includes the configuration parameters needed for the Platform Reliability Monitor detectors. Here organized by events.

SOH_ShutDownDelay

Parameter Name	SUM_SOH_Shutdown_Thresh_Init
Description	Initial value of SUM_SOH_Shutdown_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ShutDown_Timer_Thresh_Init
Description	Initial value of SUM_SOH_ShutDown_Timer_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	60

Parameter Name	SOH_NvMShutdownCntr_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMShutdownDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F091_ShutdownDID_Reference
Description	Reference to DCM Shutdown DID counter Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

4 Runtime View (Dynamic View)

4.1 Module Initialization

[$SDD_SOH 001]
The SOH implements an init function, `SOH_Init()`, which can be called either from a BswM action list during startup or from a runnable Runnable_SOH_Init.
[$Satisfies $SRS_SOH 001] [$Satisfies $SRS_SOH 007]

[$SDD_SOH 037]
If the configuration parameter *SohGenerateInitRunnable* is TRUE, the **Runnable_SOH_Init** executes `SOH_Init()` function only once, otherwise the user shall call `SOH_Init()` in the BswM initialization action list.
[$Satisfies $SRS_SOH 259]

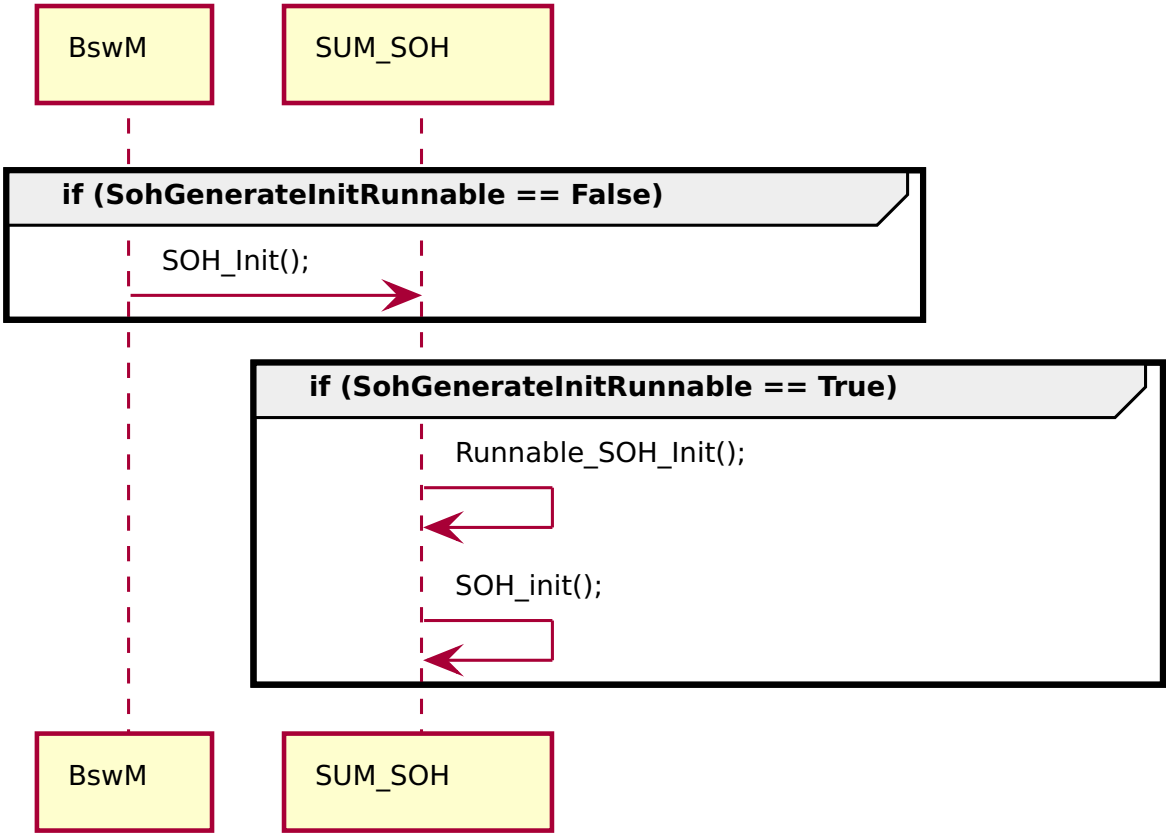


Diagram 4.1: Sequence diagram illustrating the two possible paths of initialization for the SUM_SOH module.

[$SDD_SOH 002]
The `SOH_Init` recovers all stored values from the NVM.
[$Satisfies $SRS_SOH 067] [$Satisfies $SRS_SOH 068]

[$SDD_SOH 003]
It also initializes the following groups of variables to their respective initialization values:

- SOH Main FSM init mode
- Detection counters of threshold filtering
- Detection counters of time window filtering
- Time windows of time window filtering
- DID counters of threshold filtering
- DID counters of time window filtering

[$Satisfies $SRS_SOH 023] [$Satisfies $SRS_SOH 030] [$Satisfies $SRS_SOH 042] [$Satisfies $SRS_SOH 048] [$Satisfies $SRS_SOH 055]

4.2 SOH Main

[$SDD_SOH 004]
SUM_SOH has a runnable called `Runnable_SOH_Main` that implements its SOH Main FSM (Finite State Machine) with an Init, Normal and Shutdown modes that will be used to control the functional execution of SUM behaviors relative to the appropriate state.
`Runnable_SOH_Main` is scheduled periodicly accordingly with the frequency set in the calibration parameter `SOH_RunnableSOHMainPeriod`.
[$Satisfies $SRS_SOH 005] [$Satisfies $SRS_SOH 010]

[$SDD_SOH 093]

- It starts the other runnables
- It sets the needed variables to their default value checking the MEC (it also asks the ERRH to start the cleaning of the NvM locations related to DID F094)
- It checks if any NvM writes are ongoing before the shutdown (including those due to clearing the DID F094 metrics, started when RID 0x02C0 is invoked and performed by the ERRH).
- Notifies the checkpoint reaching to WdgM

[$Satisfies $SRS_SOH 263] [$Satisfies $SRS_SOH 024] [$Satisfies $SRS_SOH 025]

4.2.1 SOH Main FSM

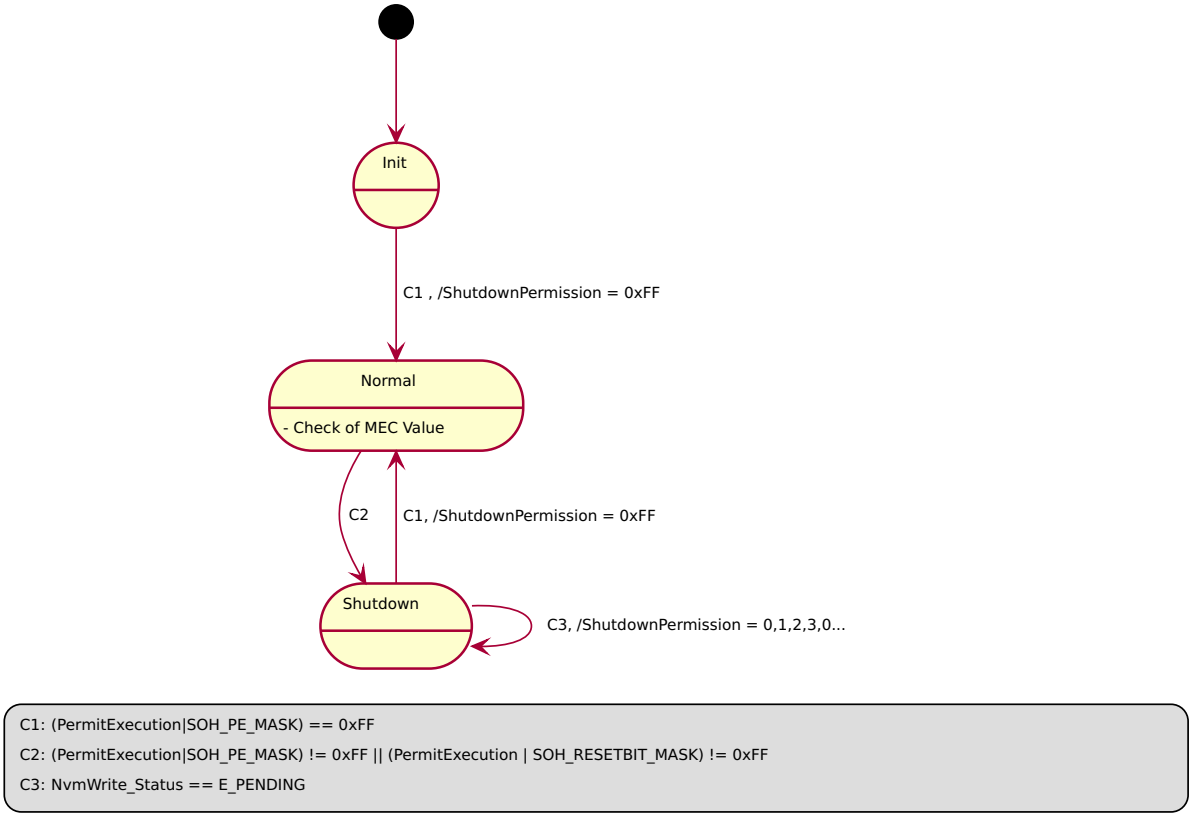


Diagram 4.2: SOH Main FSM

Init Mode

[$SDD_SOH 006]
SUM_SOH shall remains in the Init state until EcuM has reached ECUM_STATE_RUN State and all necessary AUTOSAR BSW are initialized, HW peripherals are available. The `SUSD_SOH_PermitExecution` interface is evaluated for transitionins from Init mode to other modes.
[$Satisfies $SRS_SOH 006] [$Satisfies $SRS_SOH 017]

[$SDD_SOH 007]
SUM_SOH uses the configuration SOH_PermitExection_Mask in its logic to allow configuration of which PermitExectution bits are considered to trigger a start of its functions. SUM_SOH starts the execution of its functions and moves to the Normal Mode when the Wakeup(s) Active bit in the SUSD_SOH_PermitExecution interface is set to TRUE.
[$Satisfies $SRS_SOH 018] [$Satisfies $SRS_SOH 019] [$Satisfies $SRS_SOH 257]

Normal Mode

[$SDD_SOH 008]
In Normal Mode execution of SUM_SOH should use polling for any transition criteria applicable for shutdown process to be initiated (Shutdown/Reset request in SUSD_SOH_PermitExecution.
Also ShutdownPermission interface is used to allow SUM_SUSD for continued execution to complete pending activities.
[$Satisfies $SRS_SOH 009] [$Satisfies $SRS_SOH 258]

[$SDD_SOH 009]
SUM_SOH sets its metrics and DIDs counters to their default values when the value of the (Manufacturing Enable Counter) MEC is set to 0x00
[$Satisfies $SRS_SOH 021] [$Satisfies $SRS_SOH 248]

Shutdown Mode

[$SDD_SOH 010]
SUM_SOH reads SUSD_SOH_PermitExecution and calls the monitor related to Shutdown/reset logging
[$Satisfies $SRS_SOH 020]

[$SDD_SOH 012]
SUM_SOH uses the ShutdownPermission interface to SUM_SUSD to allow for continued execution to complete pending activities (e.g.: NvM write).
[$Satisfies $SRS_SOH 008] [$Satisfies $SRS_SOH 258]

4.3 SOH Time management

This feature is described in the Architecture Design document.

4.4 Tasks management

4.4.1 Periodic Runnables

For all the Periodic Runnables definitions please refer to the documentation.

4.4.2 Event-Triggered Runnables

Operation-Invoked Event Runnables

For all the Operation-Invoked Event Runnables definitions please refer to the documentation.

Data-Receive Event Runnables

For all the Data-Receive Event Runnables definitions please refer to the documentation.

4.5 Failure handling strategy

4.6 Critical timing constraints

4.7 Runtime Scenario

4.7.1 Definitions

For all the module definitions please refer to the Doxygen documentation.

4.7.2 Variables

For all the module variables please refer to the Doxygen documentation.

4.7.3 Functions

For all the module functions please refer to the Doxygen documentation.

5 Deployment View

5.1 Resource Limits

N/A